ALGEBRAIC DE RHAM COHOMOLOGY

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ABSTRACT. In this note, we introduce several topics concerning algebraic de Rham cohomology over a smooth complex variety X.

CONTENTS

0. Notations

(1) X always denotes a complex manifold or a smooth complex algebraic variety.
(2) $\mathcal{F}, \mathcal{G}$ always denote sheaves.
(3) $\mathcal{L}$ always denotes an invertible sheaf.
(4) $\mathcal{V}$ always denotes a local system.
(5) $\mathcal{E}$ always denotes a (quasi)-coherent sheaf or a locally free sheaf.
(6) $\mathcal{I}$ always denotes an injective sheaf or an acyclic sheaf.

1. Introduction

1.1. **Comparison theorems.** At the intersection of topology and geometry, there are so-called comparison theorems, which explain that cohomology groups defined in different settings in fact reflect the same information about the manifold.

Let X be a smooth manifold. In topology, we may consider its singular cohomology $H^*_{sing}(X,\mathbb{C})$, which is given by the cohomology of the singular cochain complex with $\mathbb{C}$ coefficients. On the other hand, in differential geometry, we can use de Rham theory to define its de Rham cohomology $H^*_{dR}(X)$, which is the cohomology of the following complex:

$$0 \to \Gamma(X,\Omega^0_X) \xrightarrow{\mathrm{d}} \Gamma(X,\Omega^1_X) \xrightarrow{\mathrm{d}} \dots \xrightarrow{\mathrm{d}} \Gamma(X,\Omega^n_X) \to 0,$$

where $\Gamma(X,\Omega^k_X)$ is the $\mathbb{C}$-vector space of differential k-forms on X, and n is the dimension of X.

The classical comparison theorem says that the singular cohomology of X is isomorphic to its de Rham cohomology.

Theorem 1.1.1. Let X be a smooth manifold. Then there is an isomorphism

$$H^*_{sing}(X,\mathbb{C}) \cong H^*_{dR}(X).$$

The proof of the above comparison theorem is based on sheaf theory. Given any sheaf $\mathcal{F}$ on a smooth manifold X, we may choose an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$. Then the sheaf cohomology is defined by

$$H^*(X,\mathcal{F}) := H^*(\Gamma(X,\mathcal{I}^\bullet)).$$

In this setting, the classical Poincaré lemma can be rephrased as the following sequence of sheaves

$$0 \to \underline{\mathbb{C}} \to \Omega^0_X \xrightarrow{\mathrm{d}} \Omega^1_X \xrightarrow{\mathrm{d}} \dots \xrightarrow{\mathrm{d}} \Omega^n_X \to 0$$

is exact, and thus gives a fine resolution of the constant sheaf $\underline{\mathbb{C}}$.

In other words, in the sheaf-theoretic setting, de Rham cohomology is exactly the sheaf cohomology of the constant sheaf $\underline{\mathbb{C}}$. Similarly, the singular cochain complex also gives[1] a resolution that can be used to compute the cohomology of the constant sheaf $\underline{\mathbb{C}}$. Thus sheaf cohomology builds a bridge between singular cohomology and de Rham cohomology.

In the algebraic setting, there is also an algebraic version of de Rham complex. Let X be a smooth complex variety (with Zariski topology). The algebraic de Rham complex $\Omega^\bullet_X$ is defined to be the complex

[1] A good reference is [Wel80].

of sheaves of regular differentials as follows

$$\mathcal{O}_X \xrightarrow{\mathrm{d}} \Omega^1_X \xrightarrow{\mathrm{d}} \dots \xrightarrow{\mathrm{d}} \Omega^n_X \to 0.$$

On the other hand, X has an underlying structure of a complex manifold, denoted by X^{an}. It is therefore natural to expect a comparison theorem, but in general

$$H^*_{sing}(X^{an}, \mathbb{C}) \ncong H^*(\Gamma(X, \Omega^\bullet_X)).$$

To obtain the "right" comparison theorem, we need to consider the hypercohomology of the complex instead of simply taking global sections and then taking cohomology.

The first goal of this note is to introduce the following comparison theorem and its local system valued version.

Theorem 1.1.2 ([Gro66a])**.** Let X be a smooth complex variety and let X^{an} be the underlying complex manifold. Then there is an isomorphism

$$H^*(X^{an}, \mathbb{C}) \cong \mathbb{H}^*(X, \Omega^\bullet_X).$$

Theorem 1.1.3 ([Del70])**.** Let X be a smooth complex variety and let X^{an} be the underlying complex manifold. Let $\mathcal{E}$ be a vector bundle on X equipped with a regular integrable connection ∇, and let $\mathcal{V} = \mathcal{E}^{\nabla=0}$ be the local system of horizontal sections on the underlying complex manifold X^{an}. Then there is an isomorphism

$$H^*(X^{an}, \mathcal{V}) \cong \mathbb{H}^*(X, \Omega^\bullet_X(\mathcal{E})).$$

1.2. **Hodge to de Rham spectral sequences.** In differential geometry, we usually do not study the cohomology of the sheaf of differential k-forms, since these are fine sheaves and hence have no higher cohomology. In complex geometry, however, things become more rigid, and we also study the cohomology of sheaves of holomorphic p-forms, denoted by Ω^p_X. The holomorphic Poincaré lemma implies that the following complex

$$0 \to \Omega^p_X \to \Omega^{p,0}_X \xrightarrow{\bar{\partial}} \Omega^{p,1}_X \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} \Omega^{p,n}_X \to 0$$

is a fine resolution of Ω^p_X, and thus the Dolbeault cohomology $H^{p,q}(X) := H^q(\Gamma(X, \Omega^{p,\bullet}_X))$ computes the sheaf cohomology of Ω^p_X.

One of the most important results in complex geometry is the following Hodge decomposition theorem, which relates topological information to holomorphic information.

Theorem 1.2.1 (Hodge)**.** Let (X,ω) be a compact Kähler manifold. Then there is a decomposition

$$H^k_{dR}(X)\cong\bigoplus_{p+q=k}H^{p,q}(X).$$

A natural question is whether there is an analogue of the Hodge decomposition in algebraic geometry. The answer is affirmative: this is the E_1-degeneration of the Hodge-to-de Rham spectral sequence that we are going to introduce.

Let X be a smooth complex variety of dimension n. The Hodge filtration on the algebraic de Rham complex $\Omega_X^\bullet$ is given by

$$\Omega_X^\bullet=F^0\Omega_X^\bullet\supset F^1\Omega_X^\bullet\supset\cdots\supset F^n\Omega_X^\bullet\supset\{0\},$$

where

$$F^p\Omega_X^\bullet: 0\to\cdots\to 0\to\Omega_X^p\to\cdots\to\Omega_X^n.$$

By the standard theory of spectral sequences, this filtration gives a spectral sequence, called the Hodge-to-de Rham spectral sequence.

Theorem 1.2.2 (E_1-degeneration)**.** Let X be a smooth projective complex variety. The Hodge-to-de Rham spectral sequence attached to the algebraic de Rham complex

$$E_1^{p,q}=H^q(X,\Omega_X^p)\Longrightarrow\mathbb{H}^{p+q}(X,\Omega_X^\bullet)$$

degenerates at the E_1-page.

It is clear that the Hodge decomposition theorem implies the E_1-degeneration of the Hodge-to-de Rham spectral sequence, since every projective manifold is Kähler. However, for a long time no algebraic proof of E_1-degeneration was known. This changed with the work of P. Deligne and L. Illusie in positive characteristic, and standard degeneration arguments then allow one to deduce the degeneration of the Hodge-to-de Rham spectral sequence in characteristic zero.

Theorem 1.2.3 ([DI87])**.** Let k be an algebraically closed field of characteristic $p>0$. Let X/k be a smooth proper variety that is $W_2(k)$-liftable and of dimension $<p$. Then the Hodge-to-de Rham spectral sequence attached to the algebraic de Rham complex degenerates at the E_1-page.

Part 1. Preliminaries

In this part, we collect some basic definitions and techniques that may be used later. It is independent of our main topics, so feel free to skip it and return when you need a particular property.

2. Spectral sequence

In this section, we always assume $\mathcal{C}$ is an abelian category.

Definition 2.1. A **double complex** $K^{\bullet\bullet}$ consists of objects $K^{p,q} \in \mathcal{C}$ for $(p,q) \in \mathbb{Z}^2$, together with two sorts of differentials

(1) The horizontal differential δ, which maps $K^{p,q}$ to $K^{p+1,q}$.

(2) The vertical differential d, which maps $K^{p,q}$ to $K^{p,q+1}$.

such that $\delta^2 = \mathrm{d}^2 = 0$ and $\mathrm{d} \circ \delta = \delta \circ \mathrm{d}$.

Definition 2.2. A **morphism** $\phi \colon K^{\bullet\bullet} \to L^{\bullet\bullet}$ of double complexes consists of homomorphisms $\phi^{p,q} \colon K^{p,q} \to L^{p,q}$ which are compatible with differentials.

Definition 2.3. Let $K^{\bullet\bullet}$ be a double complex. Then the **total complex** $(K^\bullet, D)$ of $K^{\bullet\bullet}$ is defined by

$$K^n = \bigoplus_{p+q=n} K^{p,q},$$

where $D = \delta + (-1)^p \mathrm{d}$.

Definition 2.4. Let $K^{\bullet\bullet}$ be a double complex and let $K^\bullet$ be its total complex. The **first filtration** $F_H^\bullet$ is given by the subcomplexes $F_H^p = \bigoplus_{i \geq p} K^{i,j}$, with degree n component $\bigoplus_{i \geq p} K^{i,n-i}$.

Definition 2.5. Let $K^{\bullet\bullet}$ be a double complex and let $K^\bullet$ be its total complex. The **second filtration** $F_V^\bullet$ is given by the subcomplexes $F_V^p = \bigoplus_{j \geq p} K^{i,j}$, with degree n component $\bigoplus_{j \geq p} K^{n-j,j}$.

Let $(K^{\bullet\bullet}, \delta, \mathrm{d})$ be a double complex and let $(K^\bullet, D)$ be its total complex equipped with a filtration F. For any integer $r \geq 1$, we define a new filtration

$$\cdots \subseteq Z_r^{p+1} \subseteq Z_r^p \subseteq \ldots,$$

where $Z_r^p = \{x \in F^p \mid Dx \in F^{p+r}\}$, and the $(p+q)$-component of Z_r^p is denoted by $Z_r^{p,q}$. The E_r-page of the **spectral sequence** is defined to be

$$E_r^{p,q} = \frac{Z_r^{p,q}}{F^{p+1}K^{p+q} + D(F^{p-r+1}K^{p+q-1})}.$$

Now let's construct the differential

$$\mathrm{d}_r \colon E_r^{p,q} \to E_r^{p+r,q-r+1}.$$

Let $a \in E_r^{p,q}$ be represented by some $x \in F^p$ such that $Dx \in F^{p+r}$. Then $Dx \in F^{p+r}$ represents $\mathrm{d}_r(a)$. To see that this is independent of the choice of x, it suffices to show that for $x \in F^{p+1}K^{p+q}$, the class of Dx in $E_r^{p+r,q-r+1}$ is trivial. This is clear, since $Dx \in D(F^{p+1}K^{p+q})$, which is zero in $E_r^{p+r,q-r+1}$.

Remark 2.1. The beginning terms of a spectral sequence are easy to understand, namely,

$$E_1^{p,q} = H^{p+q}(F^p/F^{p+1}),$$

and the differential $\mathrm{d}_1\colon H^{p+q}(F^p/F^{p+1}) \to H^{p+q+1}(F^{p+1}/F^{p+2})$ is the connecting homomorphism in the exact sequence of complexes

$$0 \to F^{p+1}/F^{p+2} \to F^p/F^{p+2} \to F^p/F^{p+1} \to 0.$$

Proposition 2.1. The maps $\{\mathrm{d}_r\}$ satisfy $\mathrm{d}_r^2 = 0$, and the cohomology

$$\ker(\mathrm{d}_r\colon E_r^{p,q} \to E_r^{p+r,q-r+1})/\mathrm{im}(\mathrm{d}_r\colon E_r^{p-r,q+r-1} \to E_r^{p,q})$$

identifies canonically with $E_{r+1}^{p,q}$.

Proof. It is clear that $\mathrm{d}_r^2 = 0$ since $D^2 = 0$. Now let us describe the group

$$A = \ker(\mathrm{d}_r\colon E_r^{p,q} \to E_r^{p+r,q-r+1}).$$

It consists of classes in $E_r^{p,q}$ represented by some $x \in F^pK^{p+q}$ such that Dx is of the form $y + Dz$, where $y \in F^{p+r+1}K^{p+r+q}$ and $z \in F^{p+1}K^{p+q}$.

Since the class of z in $E_r^{p,q}$ is zero, the class of x is also represented by $x - z$, which satisfies $D(x - z) = y \in F^{p+r+1}$. This gives a well-defined map $f\colon A \to E_{r+1}^{p,q}$, sending the class of x to the class of $x - z$; this map is also surjective.

The kernel of f consists of the classes of $x \in F^pK^{p+q}$ such that $x - z$ is of the form $u + Dv$, where $u \in F^{p+1}K^{p+q}$ and $v \in F^{p-r}K^{p+q-1}$. In other words, one has

$$x - z - u = Dv.$$

Since the class of x is also represented by $x - z - u$, this shows $[x] \in E_r^{p,q}$ is equal to $\mathrm{d}_r([v])$, and thus the kernel of f is contained in the image of d_r. The same argument proves the inverse inclusion, which completes the proof. □

Proposition 2.2. If the filtration F of $K^\bullet$ satisfies the following condition: for every n, there exist $m_n \leq q_n$ such that $K^n \cap F^{q_n} = 0$ and $K^n \subseteq F^{m_n}$, then the spectral sequence converges to $H_D^*(K^\bullet)$, that is,

$$E_\infty^{p,q} = F^pH^{p+q}(K^\bullet).$$

Proof. Proposition 1.2.2 in [Bry08]. □

Definition 2.6. A spectral sequence **degenerates** at E_r if $E_r = E_{r+1} = \cdots = E_\infty$.

Definition 2.7. Let $E_r^{\bullet\bullet}$ and $(E'_r)^{\bullet\bullet}$ be two spectral sequences. A **morphism of spectral sequences** means that for each r, there is a homomorphism $\phi_r : E_r^{p,q} \to (E')_r^{p,q}$ such that $\mathrm{d}_r \circ \phi_r = \phi_r \circ \mathrm{d}_r$

Proposition 2.3. Let $E_r^{\bullet\bullet}$ and $(E'_r)^{\bullet\bullet}$ be convergent spectral sequences and $\phi_r : E_r^{p,q} \to (E')_r^{p,q}$ be a morphism of spectral sequences. If for some s, the morphism ϕ_s is an isomorphism, then ϕ_r is an isomorphism for all $r \geq s$, including $r = \infty$.

Proof. Proposition 1.2.4 in [Bry08]. □

3. Sheaf and its cohomology

Along this section, X denotes a topological space unless otherwise specified.

3.1. **Sheaves.**

3.1.1. *Sheaves and their morphisms.*

Definition 3.1.1. A **presheaf** of abelian groups $\mathcal{F}$ on X consists of the following data:

(1) For any open subset U of X, $\mathcal{F}(U)$ is an abelian group.
(2) If $V \subseteq U$ are two open subsets of X, then there is a group homomorphism $r_{UV}: \mathcal{F}(U) \to \mathcal{F}(V)$. Moreover, the above data satisfy
    I $\mathcal{F}(\varnothing) = 0$.
    II $r_{UU} = \mathrm{id}$.
    III If $W \subseteq V \subseteq U$ are open subsets of X, then $r_{UW} = r_{VW} \circ r_{UV}$.

Moreover, $\mathcal{F}$ is called a sheaf if it satisfies the following extra conditions

IV Let $\{V_i\}_{i\in I}$ be an open cover of an open subset $U \subseteq X$ and let $s \in \mathcal{F}(U)$. If $s|_{V_i} := r_{UV_i}(s) = 0$ for all $i \in I$, then $s = 0$.
V Let $\{V_i\}_{i\in I}$ be an open cover of an open subset $U \subseteq X$ and let $s_i \in \mathcal{F}(V_i)$. If $s_i|_{V_i\cap V_j} = s_j|_{V_i\cap V_j}$ for all $i, j \in I$, then there exists $s \in \mathcal{F}(U)$ such that $s|_{V_i} = s_i$ for all $i \in I$.

Example 3.1.1. For an abelian group G, the **constant presheaf** assigns to each open subset U the group G itself, but in general it is not a sheaf.

Definition 3.1.2. A **morphism** $\varphi: \mathcal{F} \to \mathcal{G}$ between presheaves consists of the following data:

(1) For any open subset U of X, there is a group homomorphism $\varphi(U): \mathcal{F}(U) \to \mathcal{G}(U)$.
(2) If $U \subseteq V$ are two open subsets of X, then the following diagram commutes

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, "\varphi(U)"] \arrow[d, "r_{UV}"] & \mathcal{G}(U) \arrow[d, "r_{UV}"] \\ \mathcal{F}(V) \arrow[r, "\varphi(V)"] & \mathcal{G}(V). \end{tikzcd}$$

Notation 3.1.1. For convenience, for $s \in \mathcal{F}(U)$, we often write $\varphi(s)$ instead of $\varphi(U)(s)$.

Remark 3.1.1. The morphisms between sheaves are defined as morphisms of presheaves.

Definition 3.1.3. A morphism of presheaves $\varphi\colon \mathcal{F} \to \mathcal{G}$ is called an **isomorphism** if it has a two-sided inverse; that is, there exists a morphism of presheaves $\psi\colon \mathcal{G} \to \mathcal{F}$ such that $\psi \circ \varphi = \mathrm{id}_{\mathcal{F}}$ and $\varphi \circ \psi = \mathrm{id}_{\mathcal{G}}$.

Remark 3.1.2. A morphism of presheaves $\varphi\colon \mathcal{F} \to \mathcal{G}$ is an isomorphism if and only if for every open subset $U \subseteq X$, $\varphi(U)\colon \mathcal{F}(U) \to \mathcal{G}(U)$ is an isomorphism of abelian groups.

3.1.2. *Stalks.*

Definition 3.1.4. For a presheaf $\mathcal{F}$ and $p \in X$, the **stalk** at p is defined as

$$\mathcal{F}_p = \varinjlim_{p \in U} \mathcal{F}(U)$$

Remark 3.1.3. To avoid the language of direct limits, we give a more useful but equivalent description of the stalk: for $p \in U \cap V$, sections $s_U \in \mathcal{F}(U)$ and $s_V \in \mathcal{F}(V)$ are equivalent if there exists $p \in W \subseteq U \cap V$ such that $s_U|_W = s_V|_W$. An element $s_p \in \mathcal{F}_p$, called a germ, is an equivalence class $[s_U]$.

Notation 3.1.2.

(1) For $s \in \mathcal{F}(U)$ and $p \in U$, $s|_p$ denotes the equivalent class it gives.
(2) For $s_p \in \mathcal{F}_p$, $s \in \mathcal{F}(U)$ denotes the section such that $s|_p = s_p$.

Definition 3.1.5. Given a morphism of sheaves $\varphi\colon \mathcal{F} \to \mathcal{G}$, it induces a **morphism of stalks** $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ as follows:

$$\begin{aligned} \varphi_p : \mathcal{F}_p &\to \mathcal{G}_p \\ s_p &\mapsto \varphi(s)|_p. \end{aligned}$$

Remark 3.1.4. It is necessary to check that φ_p is well-defined, since there may be different choices of s such that $s|_p = s_p$.

Proposition 3.1.1. Let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a morphism between sheaves. Then φ is an isomorphism if and only if the induced map $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is an isomorphism for every $p \in X$.

Proof. It is clear that if φ is an isomorphism between sheaves, then it induces an isomorphism between stalks. Conversely, it suffices to show that $\varphi(U)\colon \mathcal{F}(U) \to \mathcal{G}(U)$ is an isomorphism for every open subset $U \subseteq X$.

(1) Injectivity: For $s, s' \in \mathcal{F}(U)$ such that $\varphi(s) = \varphi(s')$, passing to stalks gives $\varphi_p(s|_p) = \varphi_p(s'|_p)$ for every $p \in U$, and thus $s|_p = s'|_p$ since φ_p is an isomorphism. By the definition of stalks, there exists an open subset $V_p \subseteq U$ containing p such that s agrees with s' on V_p. These sets form an open cover $\{V_p\}$ of U, and by axiom (IV) one has $s = s'$ on U.

(2) Surjectivity: For $t \in \mathcal{G}(U)$, passing to stalks shows that for every $p \in U$ there exists $s_p \in \mathcal{F}_p$ such that $\varphi_p(s_p) = t|_p$, since φ_p is surjective. By the definition of stalks, there exists an open subset $V_p \subseteq U$ containing p and a section $s \in \mathcal{F}(V_p)$ such that $\varphi(s) = t$ on V_p. This gives a collection of sections defined on an open cover $\{V_p\}$ of U, and by the injectivity proved above these sections agree on intersections. Then by axiom (V) there exists a section $s \in \mathcal{F}(U)$ such that $\varphi(s) = t$.

□

3.1.3. *Sheafification.* In Example 3.1.1, we come across a presheaf that is not a sheaf. To obtain a sheaf from a presheaf, we require a process known as sheafification. One approach to defining sheafification is through its universal property.

Definition 3.1.6. Given a presheaf $\mathcal{F}$, there is a sheaf $\mathcal{F}^+$ and a morphism $\theta\colon \mathcal{F} \to \mathcal{F}^+$, called the **sheafification**, with the property that for any sheaf $\mathcal{G}$ and any morphism $\varphi\colon \mathcal{F} \to \mathcal{G}$ there is a unique morphism $\overline{\varphi}\colon \mathcal{F}^+ \to \mathcal{G}$ such that the following diagram commutes:

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\varphi"] \arrow[d, "\theta"] & \mathcal{G} \\ \mathcal{F}^+ \arrow[ur, "\overline{\varphi}"] & \end{tikzcd}$$

The universal property shows that if the sheafification exists, then it's unique up to a unique isomorphism. One way to give an explicit construction of sheafification is to glue stalks together in a suitable way. Let $\mathcal{F}^+(U)$ be a set of functions

$$f : U \to \coprod_{p \in U} \mathcal{F}_p$$

such that $f(p) \in \mathcal{F}_p$ and for every $p \in U$ there is an open subset $V_p \subseteq U$ containing p and $t \in \mathcal{F}(V_p)$ such that $t|_q = f(q)$ for all $q \in V_p$.

Proposition 3.1.2. $\mathcal{F}^+$ is the sheafification of $\mathcal{F}$.

Proof. First let us show that $\mathcal{F}^+$ is a sheaf. It is clear that $\mathcal{F}^+$ is a presheaf, so it suffices to check conditions (IV) and (V) in the definition. Let $U \subseteq X$ be an open subset and let $\{V_i\}$ be an open cover of U.

(1) If $s \in \mathcal{F}^+(U)$ satisfies $s|_{V_i} = 0$ for all i, then s must be zero: it suffices to show $s(p) = 0$ for all $p \in U$. For any $p \in U$, there exists an open subset V_i containing p, hence $s(p) = s|_{V_i}(p) = 0$.
(2) Suppose there exists a collection of sections $\{s_i \in \mathcal{F}^+(V_i)\}_{i \in I}$ such that

$$s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$$

holds for all $i,j \in I$. Now we construct $s \in \mathcal{F}^+(U)$ as follows: For $p \in U$ and V_i containing p, we define $s(p) = s_i(p)$. This is well-defined since s_i agree on the intersections, so it remains to show $s \in \mathcal{F}^+(U)$. It's clear $s(p) \in \mathcal{F}_p$. For $p \in U$, there exists V_i containing p, and thus there exists $W_i \subseteq V_i$ containing p and $t \in \mathcal{F}(W_i)$ such that $t|_q = s_i(q) = s(q)$ for all $q \in V_p$.

There is a canonical morphism $\theta\colon \mathcal{F} \to \mathcal{F}^+$ as follows: For open subset $U \subseteq X$, and $s \in \mathcal{F}(U)$, $\theta(s)$ is defined by

$$\begin{aligned} \theta(s)\colon U &\to \coprod_{p\in U} \mathcal{F}_p \\ p &\mapsto s|_p. \end{aligned}$$

Note that if $\mathcal{F}$ is a sheaf, the canonical morphism $\theta\colon \mathcal{F} \to \mathcal{F}^+$ is an isomorphism.

(1) Injectivity: If $s \in \mathcal{F}(U)$ satisfies $s|_p = 0$ for all $p \in U$, then there exists an open cover $\{V_i\}_{i\in I}$ of U such that $s|_{V_i} = 0$; by axiom (IV) of sheaves, one has $s = 0$.
(2) Surjectivity: For $f \in \mathcal{F}^+(U)$ and $p \in U$, there exists $p \in V_p \subseteq U$ and $t \in \mathcal{F}(V_p)$ such that $f(p) = t|_p$ by construction of $\mathcal{F}^+$. Then glue these sections together to get our desired s such that $\theta(s) = f$.

Finally, let us show that $\mathcal{F}^+$ satisfies the universal property of sheafification. A morphism of presheaves $\varphi\colon \mathcal{F} \to \mathcal{G}$ induces a map on stalks

$$\varphi_p\colon \mathcal{F}_p \to \mathcal{G}_p.$$

For $f \in \mathcal{F}^+(U)$, the composite of f with the map

$$\coprod_{p\in U} \varphi_p\colon \coprod_{p\in U} \mathcal{F}_p \to \coprod_{p\in U} \mathcal{G}_p$$

gives a map $\widetilde{\varphi}(f)\colon U \to \coprod_{p\in U} \mathcal{G}_p$, and in fact $\widetilde{\varphi}(f) \in \mathcal{G}^+(U)$: For $p \in U$, $\widetilde{\varphi}(f)(p) \in \mathcal{G}_p$ since $f(p) \in \mathcal{F}_p$ and $\varphi_p\colon \mathcal{F}_p \to \mathcal{G}_p$. If for all $q \in V_p$ we have $t|_q = f(q)$, then

$$\widetilde{\varphi}(f)(q) = \varphi_q(f(q)) = \varphi_q(t|_q) = \varphi(t)|_q.$$

Since $\mathcal{G}$ is a sheaf, the canonical morphism $\theta'\colon \mathcal{G} \to \mathcal{G}^+$ is an isomorphism, and so we can define $\overline{\varphi} := \theta'^{-1} \circ \widetilde{\varphi}$. Now let's show $\varphi = \overline{\varphi} \circ \theta = \theta'^{-1} \circ \widetilde{\varphi} \circ \theta$. It's easy to show they coincide on each stalk since $\varphi_p = \theta_p'^{-1} \circ \widetilde{\varphi}_p \circ \theta_p$, and thus $\varphi = \overline{\varphi} \circ \theta$ by Proposition 3.1.1. Furthermore, uniqueness follows from the fact that $\overline{\varphi}_p$ is uniquely determined by φ_p. □

Remark 3.1.5. From the construction, one can see the stalk of $\mathcal{F}^+$ at p is exactly $\mathcal{F}_p$.

Remark 3.1.6. Sheafification can be described in more categorical language: let $\underline{Ab}_X$ denote the category of sheaves of abelian groups on X, and let the category of presheaves contain $\underline{Ab}_X$ as a full subcategory. There is a natural inclusion functor ι from sheaves to presheaves. The sheafification functor is left adjoint to ι.

Example 3.1.2. For an abelian group G, the associated **constant sheaf** $\underline{G}$ is the sheafification of the constant presheaf. By the construction of sheafification, $\underline{G}$ can be explicitly expressed as

$$\underline{G}(U) = \{\text{locally constant function } f : U \to G\}$$

3.1.4. *Exact sequences of sheaves.* Given a morphism $\varphi\colon \mathcal{F} \to \mathcal{G}$ between sheaves of abelian groups, there are the following presheaves

$$\begin{aligned} U &\mapsto \ker\varphi(U) \\ U &\mapsto \operatorname{im}\varphi(U) \\ U &\mapsto \operatorname{coker}\varphi(U), \end{aligned}$$

since $\varphi(U)\colon \mathcal{F}(U) \to \mathcal{G}(U)$ is a group homomorphism.

Proposition 3.1.3. The kernel of a morphism between sheaves is a sheaf.

Proof. Let $\{V_i\}_{i\in I}$ be an open cover of U.

(1) For $s \in \ker\varphi(U)$, if $s|_{V_i} = 0$, then $s = 0$ since s is also in $\mathcal{F}(U)$.
(2) If there exists $s_i \in \ker\varphi(V_i)$ such that $s_i|_{V_i\cap V_j} = s_j|_{V_i\cap V_j}$, then they glue together to get $s \in \mathcal{F}(U)$. Note that

$$\varphi(U)(s)|_{V_i} = \varphi(V_i)(s|_{V_i}) = \varphi(V_i)(s_i) = 0$$

Then $s \in \ker\varphi(U)$.

□

But the image of a morphism may not be a sheaf. Although we can prove the first requirement in the same way, the proof of the second requirement fails: if there exists $s_i \in \operatorname{im}\varphi(V_i)$ and we can glue them together to obtain $s \in \mathcal{G}(U)$, then s may still not be the image of any $t \in \mathcal{F}(U)$. The cokernel fails to be a sheaf for the same reason.

Definition 3.1.7. Let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a morphism between sheaves of abelian groups. Then the **image** and **cokernel** of φ is defined to be the sheafification of the following presheaves

$$\begin{aligned} U &\mapsto \operatorname{im}\varphi(U) \\ U &\mapsto \operatorname{coker}\varphi(U) \end{aligned}$$

respectively.

Definition 3.1.8. For a sequence of sheaves:

$$\cdots \to \mathcal{F}^{i-1} \xrightarrow{\varphi^{i-1}} \mathcal{F}^i \xrightarrow{\varphi^i} \mathcal{F}^{i+1} \to \dots$$

It is called **exact** at $\mathcal{F}^i$ if $\ker \varphi^i = \operatorname{im} \varphi^{i-1}$. If a sequence is exact everywhere, then it is an exact sequence of sheaves.

Definition 3.1.9. An exact sequence of sheaves is called a **short exact sequence** if it's of the form

$$0 \to \mathcal{F} \xrightarrow{\varphi} \mathcal{G} \xrightarrow{\psi} \mathcal{H} \to 0.$$

Proposition 3.1.4. Let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a morphism between sheaves of abelian groups. Then for any $p \in X$, one has

$$(\ker \varphi)_p = \ker \varphi_p$$
$$(\operatorname{im} \varphi)_p = \operatorname{im} \varphi_p.$$

Proof. (1). It's clear $(\ker \varphi)_p \subseteq \ker \varphi_p$. Conversely, if $s_p \in \ker \varphi_p$, then $\varphi_p(s_p) = 0 \in \mathcal{G}_p$. In other words, there exists an open subset U containing p and $s \in \mathcal{F}(U)$ such that $s|_p = s_p$ and $\varphi(s)|_p = 0$, which implies there is another open subset V containing p such that $\varphi(s)|_V = 0$. Hence $\varphi(s|_V) = 0$, that is, $s|_V \in \ker \varphi(V)$. Thus $s_p = (s|_V)|_p \in (\ker \varphi)_p$.

(2). It is clear that $(\operatorname{im} \varphi)_p \subseteq \operatorname{im} \varphi_p$, since sheafification does not change stalks. Conversely, if $s_p \in \operatorname{im} \varphi_p$, then there exists $t_p \in \mathcal{F}_p$ such that $\varphi_p(t_p) = s_p$. Suppose $t \in \mathcal{F}(U)$ is a section on some open subset U containing p such that $t|_p = t_p$. Then $\varphi(t)|_p = \varphi_p(t_p) = s_p$. In other words, s_p is in the stalk of the image presheaf at p, and since sheafification does not change stalks, we have $s_p \in (\operatorname{im} \varphi)_p$. $\square$

Corollary 3.1.1. The sequence of sheaves

$$\cdots \to \mathcal{F}^{i-1} \xrightarrow{\varphi^{i-1}} \mathcal{F}^i \xrightarrow{\varphi^i} \mathcal{F}^{i+1} \to \dots$$

is exact if and only if the sequence of abelian groups are exact

$$\cdots \to \mathcal{F}_p^{i-1} \xrightarrow{\varphi_p^{i-1}} \mathcal{F}_p^i \xrightarrow{\varphi_p^i} \mathcal{F}_p^{i+1} \to \dots$$

for all $p \in X$.

Corollary 3.1.2. The sequence of sheaves

$$0 \to \mathcal{F} \to \mathcal{G}$$

is exact if and only if for any open subset U, the following sequence of abelian groups is exact

$$0 \to \mathcal{F}(U) \to \mathcal{G}(U).$$

Proof. For each $p \in U$, we have

$$\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$$

is injective. That is $\ker \varphi_p = 0$. So we obtain $(\ker \varphi(U))_p = 0$ for all $p \in U$. But for a section $s \in \mathcal{F}(U)$ if we have $s|_p = 0$, then we must have $s = 0$, and thus $\ker \varphi(U) = 0$. □

Example 3.1.3. Let X be a complex manifold and $\mathcal{O}_X$ be its holomorphic function sheaf. Then

$$0 \to 2\pi\sqrt{-1}\underline{\mathbb{Z}} \to \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \to 0$$

is an exact sequence of sheaves, which is called **exponential sequence**.

Proof. The difficulty is to show exponential map is surjective on stalks at $p \in X$. That is we need to construct logarithms of functions $g \in \mathcal{O}_X^*(U)$ for U, a neighborhood of p. We may choose U is simply-connected, then define

$$\log g(q) = \log g(p) + \int_{\gamma_q} \frac{\mathrm{d}g}{g}$$

for $q \in U$, where γ_q is a path from p to q in U, and the definition of $\log g(q)$ is independent of the choice of γ_q since U is simply-connected. □

Remark 3.1.7. In fact, U is simply-connected is crucial for constructing logarithm. If we consider $X = \mathbb{C}$ and $U = \mathbb{C}\setminus\{0\}$, then

$$\exp\colon \mathcal{O}_X(U) \to \mathcal{O}_X^*(U)$$

cannot be surjective.

3.1.5. *Direct image.*

Definition 3.1.10. Let $f : X \to Y$ be a continuous map between topological spaces, and let $\mathcal{F}$ be a sheaf of abelian groups on X. The **direct image** of $\mathcal{F}$, denoted by $f_*\mathcal{F}$, is a sheaf on Y defined by

$$f_*\mathcal{F}(U) := \mathcal{F}(f^{-1}(U)).$$

Proposition 3.1.5. $f_* : \underline{Ab}_X \to \underline{Ab}_Y$ is a left exact functor.

Proof. Given an exact sequence of sheaves on X

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}''.$$

Then we need to show

$$0 \to f_*\mathcal{F}' \to f_*\mathcal{F} \to f_*\mathcal{F}''$$

is also an exact sequence of sheaves on Y. By Remark **??** it suffices to show that for any open subset $V \subseteq Y$, we have the following exact sequence

$$0 \to f_*\mathcal{F}'(V) \to f_*\mathcal{F}(V) \to f_*\mathcal{F}''(V),$$

and that's exactly

$$0 \to \mathcal{F}'(f^{-1}(V)) \to \mathcal{F}(f^{-1}(V)) \to \mathcal{F}''(f^{-1}(V)).$$

Since f is continuous, $f^{-1}(V)$ is an open subset of X, and thus the above sequence of abelian groups is exact because $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}''$ is exact. □

3.2. **Sheaf cohomology.**

3.2.1. *Derived functor formulation.* Let $\underline{Ab}_X$ denote the category of sheaves of abelian groups on X. In this section we will introduce sheaf cohomology by considering it as a derived functor.

Given an exact sequence of sheaves

$$0 \to \mathcal{F}' \xrightarrow{\phi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}''.$$

By taking sections over an open subset U, we obtain a sequence of abelian groups

$$0 \to \mathcal{F}'(U) \xrightarrow{\phi(U)} \mathcal{F}(U) \xrightarrow{\psi(U)} \mathcal{F}''(U).$$

The above sequence is exact not only at $\mathcal{F}'(U)$, but also at $\mathcal{F}(U)$. In other words, the functor of taking sections over an open subset is left exact.

(1) First let us show $\ker\psi(U) \supseteq \operatorname{im}\phi(U)$. For $s \in \mathcal{F}'(U)$, to show $\psi \circ \phi(s) = 0$, it suffices to show $(\psi \circ \phi(s))|_p = 0$ for all $p \in U$, since $\mathcal{F}''$ is a sheaf. For any $p \in U$, by considering the stalk at p we obtain an exact sequence of abelian groups

$$0 \to \mathcal{F}'_p \xrightarrow{\phi_p} \mathcal{F}_p \xrightarrow{\psi_p} \mathcal{F}''_p.$$

Then we obtain $\psi_p \circ \phi_p(s|_p) = 0$, which implies $(\psi \circ \phi(s))|_p = 0$.

(2) Conversely, given $s \in \ker\psi(U)$, we have $s|_p \in \ker\psi_p$ for any $p \in U$. By exactness of stalks, there exists $t_p \in \mathcal{F}'_p$ such that $\phi_p(t_p) = s|_p$. Thus there exists an open subset V_i containing p and a section $t_i \in \mathcal{F}'(V_i)$ such that $\phi(t_i) = s|_{V_i}$. Now it suffices to show that these t_i can be glued together to obtain $t \in \mathcal{F}'(U)$. Since $\mathcal{F}'$ is a sheaf, it suffices to check that the t_i agree on intersections $V_i \cap V_j$. Note that $\phi(t_i - t_j|_{V_i \cap V_j}) = s|_{V_i \cap V_j} - s|_{V_i \cap V_j} = 0$, so these t_i agree on intersections since ϕ is injective.

In homological algebra, one often considers the derived functor of a left or right exact functor. In particular, the functor of taking global sections is left exact, and its right derived functor defines sheaf cohomology. Before giving the definition of the derived functor, let us first define an injective resolution of a sheaf.

Definition 3.2.1. A sheaf $\mathcal{I}$ is **injective** if $\mathrm{Hom}(-,\mathcal{I})$ is an exact functor.

Definition 3.2.2. Let $\mathcal{F}$ be a sheaf. An **injective resolution** of $\mathcal{F}$ is an exact sequence

$$0 \to \mathcal{F} \to \mathcal{I}^0 \to \mathcal{I}^1 \to \mathcal{I}^2 \to \dots$$

where $\mathcal{I}^i$ are injective for all i.

Theorem 3.2.1.

(1) Every sheaf admits an injective resolution.
(2) Let $\mathcal{F} \to \mathcal{I}^\bullet$ and $\mathcal{G} \to \mathcal{G}^\bullet$ be two resolutions, and let $\phi\colon \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then there exists a morphism $\widetilde{\phi}\colon \mathcal{I}^\bullet \to \mathcal{G}^\bullet$ that lifts ϕ, and it is unique up to homotopy.

Proof. Proposition 1.1.15 in [Bry08]. □

Theorem 3.2.2. Let $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{Q} \to 0$ be an exact sequence of sheaves of abelian groups. Let $\mathcal{F} \to \mathcal{I}^\bullet$ and $\mathcal{Q} \to \mathcal{J}^\bullet$ be injective resolutions. Then there exists an injective resolution $\mathcal{G} \to \mathcal{K}^\bullet$ such that the following diagram commutes

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathcal{F} \arrow[r] \arrow[d] & \mathcal{I}^0 \arrow[r] \arrow[d] & \mathcal{I}^1 \arrow[r] \arrow[d] & \dots \\
0 \arrow[r] & \mathcal{G} \arrow[r] \arrow[d] & \mathcal{K}^0 \arrow[r] \arrow[d] & \mathcal{K}^1 \arrow[r] \arrow[d] & \dots \\
0 \arrow[r] & \mathcal{Q} \arrow[r] \arrow[d] & \mathcal{J}^0 \arrow[r] \arrow[d] & \mathcal{J}^1 \arrow[r] \arrow[d] & \dots \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

where each sequence $0 \to \mathcal{I}^n \to \mathcal{K}^n \to \mathcal{J}^n \to 0$ is exact.

Proof. Proposition 1.1.18 of [Bry08]. □

Definition 3.2.3. Let $\mathcal{F}$ be a sheaf of abelian groups. Then its **cohomology**

$$H^p(X,\mathcal{F}) := H^p(\Gamma(X,\mathcal{I}^\bullet)).$$

Remark 3.2.1. Theorem 3.2.1 shows that the definition of sheaf cohomology is independent of the choice of injective resolution.

Example 3.2.1. The cohomology of zero degree consists of the global sections.

Example 3.2.2. If $\mathcal{F}$ is an injective sheaf, then $H^i(X,\mathcal{F}) = 0$ for all $i > 0$, since the sheaf cohomology of an injective sheaf can be computed using the following special injective resolution:

$$0 \to \mathcal{F} \xrightarrow{\mathrm{id}} \mathcal{F} \to 0 \to 0 \to \dots$$

Theorem 3.2.3 (zig-zag)**.** If

$$0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$$

is a short sequence of sheaves, then there is an induced long exact sequence of abelian groups

$$0 \to H^0(X,\mathcal{F}) \to H^0(X,\mathcal{G}) \to H^0(X,\mathcal{H}) \to H^1(X,\mathcal{F}) \to H^1(X,\mathcal{G}) \to \dots$$

Proof. Corollary 1.1.19 in [Bry08]. □

3.2.2. *Hypercohomology.* In homological algebra, hypercohomology generalizes the cohomology functor by taking as input not objects in an abelian category, but complexes of such objects.

Definition 3.2.4. Let $\mathcal{F}^\bullet$ be a bounded below[2] complex of sheaves on X. The **hypercohomology** of the complex $\mathcal{F}^\bullet$ is defined as the total cohomology of the double complex $\Gamma(X,\mathcal{I}^{\bullet\bullet})$, where $\mathcal{I}^{p,\bullet}$ is an injective resolution of $\mathcal{F}^p$ for each p.

In fact, we can find the following injective resolution with better properties.

Proposition 3.2.1. Let $(\mathcal{F}^\bullet, \mathrm{d}_\mathcal{F})$ be a bounded below complex of sheaves on X. There exists a double complex $(\mathcal{I}^{\bullet\bullet}, \delta, \mathrm{d})$ with $\mathcal{I}^{p,q} = 0$ for $q < 0$, and a morphism of complexes $u\colon \mathcal{F}^\bullet \to (\mathcal{I}^{\bullet,0}, \mathrm{d})$ such that

(1) For each $p \in \mathbb{Z}$, the complex of sheaves $(\mathcal{I}^{p,\bullet}, \mathrm{d})$ is an injective resolution of $\mathcal{F}^p$.
(2) For each $p \in \mathbb{Z}$, the complex of sheaves $\mathrm{im}\{\delta\colon \mathcal{I}^{p-1,\bullet} \to \mathcal{I}^{p,\bullet}\} \subseteq \mathcal{I}^{p,\bullet}$ is an injective resolution of $\mathrm{im}\{\mathrm{d}_\mathcal{F}\colon \mathcal{F}^{p-1} \to \mathcal{F}^p\}$.
(3) For each $p \in \mathbb{Z}$, the complex of sheaves $\ker\{\delta\colon \mathcal{I}^{p,\bullet} \to \mathcal{I}^{p+1,\bullet}\} \subseteq \mathcal{I}^{p,\bullet}$ is an injective resolution of $\ker\{\mathrm{d}_\mathcal{F}\colon \mathcal{F}^p \to \mathcal{F}^{p+1}\}$.
(4) For each $p \in \mathbb{Z}$, the complex of sheaves $\mathcal{H}^{p,\bullet}(\mathcal{I}^{\bullet\bullet})$ (the horizontal cohomology) is an injective resolution of $\mathcal{H}^p(\mathcal{F}^\bullet)$.

---

[2]From now on, we always assume our complex of sheaves of abelian groups is bounded below for convenience.

Furthermore, a morphism between complexes of sheaves induces a morphism of double complexes, which is unique up to homotopy.

$$
\begin{tikzcd}
 & \vdots & \vdots & \vdots & \\
\cdots \arrow[r] & \mathcal{I}^{p-1,1} \arrow[r] \arrow[u] & \mathcal{I}^{p,1} \arrow[r] \arrow[u] & \mathcal{I}^{p+1,1} \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & \mathcal{I}^{p-1,0} \arrow[r] \arrow[u] & \mathcal{I}^{p,0} \arrow[r] \arrow[u] & \mathcal{I}^{p+1,0} \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & \mathcal{F}^{p-1} \arrow[r] \arrow[u] & \mathcal{F}^{p} \arrow[r] \arrow[u] & \mathcal{F}^{p+1} \arrow[r] \arrow[u] & \cdots
\end{tikzcd}
$$

Proof. Since $\mathcal{F}^\bullet$ is bounded from below, we may choose k such that $\mathcal{F}^p = 0$ for $p < k$. First, we construct an injective resolution $\mathcal{H}^{p,\bullet}$ of $\mathcal{H}^p(\mathcal{F}^\bullet)$ for all p, and an injective resolution $\mathcal{R}^{\bullet,p}$ for $\operatorname{im}\{\mathrm{d}_{\mathcal{F}}\colon \mathcal{F}^{p-1} \to \mathcal{F}^p\}$. Then by Theorem 3.2.2 there is an injective resolution $\mathcal{S}^{p,\bullet}$ of $\ker\{\mathrm{d}_{\mathcal{F}}\colon \mathcal{F}^q \to \mathcal{F}^{q+1}\}$ such that the following diagram commutes:

$$
\begin{tikzcd}
0 \arrow[r] & \mathcal{R}^{p,\bullet} \arrow[r] & \mathcal{S}^{p,\bullet} \arrow[r] & \mathcal{H}^{p,\bullet} \arrow[r] & 0 \\
0 \arrow[r] & \operatorname{im}\{\mathrm{d}_{\mathcal{F}}\colon \mathcal{F}^{p-1} \to \mathcal{F}^p\} \arrow[r] \arrow[u] & \ker\{\mathrm{d}_{\mathcal{F}}\colon \mathcal{F}^q \to \mathcal{F}^{q+1}\} \arrow[r] \arrow[u] & \mathcal{H}^p(\mathcal{F}^\bullet) \arrow[r] \arrow[u] & 0.
\end{tikzcd}
$$

Then by induction on p, we may use Theorem 3.2.2 again to construct an injective resolution $\mathcal{I}^{p,\bullet}$ of $\mathcal{F}^p$ such that the following diagram commutes

$$
\begin{tikzcd}
0 \arrow[r] & \mathcal{S}^{p,\bullet} \arrow[r] & \mathcal{I}^{p,\bullet} \arrow[r] & \mathcal{R}^{p+1,\bullet} \arrow[r] & 0 \\
0 \arrow[r] & \ker\{\mathrm{d}_{\mathcal{F}}\colon \mathcal{F}^q \to \mathcal{F}^{q+1}\} \arrow[r] \arrow[u] & \mathcal{F}^p \arrow[r] \arrow[u] & \operatorname{im}\{\mathrm{d}_{\mathcal{F}}\colon \mathcal{F}^{p} \to \mathcal{F}^{p+1}\} \arrow[r] \arrow[u] & 0.
\end{tikzcd}
$$

The horizontal differential $\mathrm{d}\colon \mathcal{I}^{p,\bullet} \to \mathcal{I}^{p+1,\bullet}$ is defined to be the composition of the maps $\mathcal{I}^{p,\bullet} \to \mathcal{R}^{p+1,\bullet} \to \mathcal{S}^{p+1,\bullet} \to \mathcal{I}^{p+1,\bullet}$. This gives the desired double complex $\mathcal{I}^{\bullet\bullet}$. □

Theorem 3.2.4. Let $\mathcal{F}^\bullet$ be a bounded below complex of sheaves on X. Then there is a spectral sequence converging to the hypercohomology $\mathbb{H}^*(X,\mathcal{F}^\bullet)$ with $E_1^{p,q} = H^q(X,\mathcal{F}^p)$. The differential $\mathrm{d}_1\colon H^q(X,\mathcal{F}^p) \to H^q(X,\mathcal{F}^{p+1})$ is induced by the morphism of sheaves $\mathcal{F}^p \to \mathcal{F}^{p+1}$.

Proof. Let $\mathcal{I}^{\bullet\bullet}$ be an injective resolution of $\mathcal{F}^\bullet$ and consider the first filtration of the double complex $\Gamma(X,\mathcal{I}^{\bullet\bullet})$. Note that the E_1-page is the vertical cohomology; that is, $E_1^{p,q}$ is the degree q cohomology of the complex $\Gamma(X,\mathcal{I}^{p,\bullet})$. Since $\mathcal{I}^{p,\bullet}$ is an injective resolution of the sheaf $\mathcal{F}^p$, we have $E_1^{p,q} = H^q(X,\mathcal{F}^p)$.

Since the complex $\mathcal{F}^\bullet$ is bounded below, Proposition 2.2 implies that the spectral sequence converges to the total cohomology of $\Gamma(X,\mathcal{I}^{\bullet\bullet})$, namely the hypercohomology $\mathbb{H}^*(X,\mathcal{F}^\bullet)$. □

In fact, hypercohomology generalizes ordinary sheaf cohomology: any sheaf $\mathcal{F}$ gives a complex of sheaves $\mathcal{F}^\bullet[0]$, and computing its hypercohomology recovers the sheaf cohomology of $\mathcal{F}$.

Definition 3.2.5. For a sheaf $\mathcal{F}$, the **shifted complex** $\mathcal{F}^\bullet[n]$ is a complex of sheaves defined by

$$(\mathcal{F}^\bullet[n])^i = \begin{cases} \mathcal{F}, & i = n \\ 0, & \text{otherwise.} \end{cases}$$

Corollary 3.2.1. Let $\mathcal{F}^\bullet$ be a complex of sheaves on X. If each sheaf in $\mathcal{F}^\bullet$ is acyclic, then there is an isomorphism between the hypercohomology $\mathbb{H}^*(X,\mathcal{F}^\bullet)$ and the cohomology groups of the complex

$$\cdots \to \Gamma(X,\mathcal{F}^{p-1}) \to \Gamma(X,\mathcal{F}^p) \to \ldots$$

In particular, the hypercohomology is a generalization of the usual cohomology.

Proof. Note that the assumption means that the spectral sequence attached to the complex $\mathcal{F}^\bullet$ satisfies $E_1^{p,q} = 0$ for $q > 0$, and thus it degenerates at E_2-page. This shows $\mathbb{H}^p(X,\mathcal{F}^\bullet) = E_2^{p,0}$, which is the cohomology groups of the complex

$$\cdots \to \Gamma(X,\mathcal{F}^{p-1}) \to \Gamma(X,\mathcal{F}^p) \to \ldots$$

as desired. □

3.3. **Čech cohomology.**

3.3.1. *Čech cohomology of presheaf.* Let $\mathcal{F}$ be a presheaf on X and let $\mathfrak{U} = \{U_\alpha\}_{\alpha\in J}$ be an open cover of X. For convenience, we use $U_{\alpha\beta}$ to denote the intersection $U_\alpha \cap U_\beta$, and similarly for $U_{\alpha\beta\gamma}$. Now consider the following complex:

$$0 \to C^0(\mathfrak{U},\mathcal{F}) \xrightarrow{\delta} C^1(\mathfrak{U},\mathcal{F}) \xrightarrow{\delta} C^2(\mathfrak{U},\mathcal{F}) \to \ldots,$$

where $C^p(\mathfrak{U},\mathcal{F})=\prod \mathcal{F}(U_{\alpha_0\dots\alpha_p})$, and the differential δ is defined by

$$\begin{aligned}\delta\colon C^p(\mathfrak{U},\mathcal{F}) &\to C^{p+1}(\mathfrak{U},\mathcal{F})\\ \omega &\mapsto \sum_{i=0}^{p+1}(-1)^i\omega_{\alpha_0\dots\widehat{\alpha_i}\dots\alpha_{p+1}}.\end{aligned}$$

A routine computation shows that $(C^p(\mathfrak{U},\mathcal{F}),\delta)$ forms a complex.

Definition 3.3.1. The **Čech cohomology** of the presheaf $\mathcal{F}$ with respect to the open cover $\mathfrak{U}$ is defined as the cohomology of the complex $(C^p(\mathfrak{U},\mathcal{F}),\delta)$.

Note that the definition of Čech cohomology depends on the choice of open cover, so it is natural to ask what happens if we consider different open covers.

Lemma 3.3.1. Given an open cover $\mathfrak{U}=\{U_\alpha\}_{\alpha\in I}$ and a refinement $\mathfrak{V}=\{V_\beta\}_{\beta\in J}$, if ϕ,ψ are two refinement maps $J\to I$, then there is a homotopy operator between $\phi^\#$ and $\psi^\#$.

Proof. Define $K\colon C^q(\mathfrak{U},\mathcal{F})\to C^{q-1}(\mathfrak{V},\mathcal{F})$ by

$$(K\omega)(V_{\beta_0\dots\beta_{q-1}})=\sum(-1)^i\omega(U_{\phi(\beta_0)\dots\phi(\beta_i)\psi(\beta_i)\dots\psi(\beta_{q-1})}).$$

Now we claim the following formula holds

$$\psi^\#-\phi^\#=\delta K+K\delta.$$

Indeed, take a cochain $\omega\in C^q(\mathfrak{U},\mathcal{F})$ and an intersection $V_{\beta_0\dots\beta_q}$ of the open cover. Then it is easy to see that

$$\psi^\#-\phi^\#(\omega)(V_{\beta_0\dots\beta_q})=\omega(U_{\psi(\beta_0)\dots\psi(\beta_q)})-\omega(U_{\phi(\beta_0)\dots\phi(\beta_q)}).$$

Now let's compute $\delta K\omega$ as follows:

$$\begin{aligned}\delta K(\omega)(V_{\beta_0\dots\beta_q}) &=\sum_i(-1)^iK\omega(V_{\beta_0\dots\widehat{\beta_i}\dots\beta_q})\\ &=\underbrace{\sum_{i\leq j}(-1)^{i+j}\omega(U_{\phi(\beta_0)\dots\widehat{\phi(\beta_i)}\dots\phi(\beta_{j+1})\psi(\beta_{j+1})\dots\psi(\beta_q)})}_{\text{part I}}\\ &\quad+\underbrace{\sum_{i>j}(-1)^{i+j}\omega(U_{\phi(\beta_0)\dots\phi(\beta_j)\psi(\beta_j)\dots\widehat{\psi(\beta_j)}\dots\psi(\beta_q)})}_{\text{part II}}.\end{aligned}$$

Similarly we have $K\delta\omega$ as follows

$$\begin{aligned}
K\delta\omega(V_{\beta_0\dots\beta_q}) &= \sum_j (-1)^j \delta\omega(U_{\phi(\beta_0)\dots\phi(\beta_j)\psi(\beta_j)\dots\psi(\beta_q)}) \\
&= \underbrace{\sum_{i<j} (-1)^{i+j} \omega(U_{\phi(\beta_0)\dots\widehat{\phi(\beta_i)}\dots\phi(\beta_j)\psi(\beta_j)\dots\psi(\beta_q)})}_{\text{part III}} \\
&\quad + \underbrace{\sum_{i>j} (-1)^{i+j} \omega(U_{\phi(\beta_0)\dots\phi(\beta_j)\psi(\beta_j)\dots\widehat{\psi(\beta_{i-1})}\dots\psi(\beta_q)})}_{\text{part IV}} \\
&\quad + \underbrace{\sum_j \omega(U_{\phi(\beta_0)\dots\widehat{\phi(\beta_j)}\psi(\beta_j)\dots\psi(\beta_q)})}_{\text{part V}}.
\end{aligned}$$

Note that part I cancels with part III, since if one fixes i, the j-th terms of part I and part III are equal but have opposite signs. Similarly, part II and part IV almost cancel each other, but

$$\text{part II} + \text{part IV} = \underbrace{\sum_j -\omega(U_{\phi(\beta_0)\dots\phi(\beta_j)\widehat{\psi(\beta_j)}\psi(\beta_{j+1})\dots\psi(\beta_q)})}_{\text{part VI}},$$

and it is clear that

$$\text{part V} + \text{part VI} = \omega(U_{\psi(\beta_0)\dots\psi(\beta_q)}) - \omega(U_{\phi(\beta_0)\dots\phi(\beta_q)})$$

as desired. This completes the proof. □

Thus for two different open coverings $\mathfrak{U}, \mathfrak{V}$ such that $\mathfrak{V}$ is a refinement of $\mathfrak{U}$, there is a natural homomorphism

$$f_{\mathfrak{U}\mathfrak{V}} : H^*(\mathfrak{U}, \mathcal{F}) \to H^*(\mathfrak{V}, \mathcal{F}).$$

Furthermore, if there are three open covers such that $\mathfrak{C}$ is a refinement of $\mathfrak{V}$ and $\mathfrak{V}$ is a refinement of $\mathfrak{U}$, then we have

$$f_{\mathfrak{U}\mathfrak{C}} = f_{\mathfrak{U}\mathfrak{V}} \circ f_{\mathfrak{V}\mathfrak{C}}.$$

If we put a partial order on the set of all open covers by declaring $\mathfrak{U} < \mathfrak{V}$ if and only if $\mathfrak{V}$ is a refinement of $\mathfrak{U}$, then $\{H^*(\mathfrak{U}, \mathcal{F}), f_{\mathfrak{U}\mathfrak{V}}\}$ is a direct system.

Definition 3.3.2. The direct limit of the direct system $\{H^*(\mathfrak{U}, \mathcal{F}), f_{\mathfrak{U}\mathfrak{V}}\}$ is called the **Čech cohomology** of X with values in the presheaf $\mathcal{F}$, that is,

$$\check{H}^*(X, \mathcal{F}) := \varinjlim_{\mathfrak{U}} H^*(\mathfrak{U}, \mathcal{F}).$$

3.3.2. *Comparison theorem for Čech cohomology.*

Proposition 3.3.1. Let $\mathcal{F}$ be a sheaf on X and let $\mathfrak{U}=\{U_i\}_{i\in I}$ be an open cover of X. For every intersection $U_{i_1\dots i_k}$, we use j to denote the inclusion $U_{i_1\dots i_k}\to X$. Then

$$0\to\mathcal{F}\to\prod_i j_*\mathcal{F}|_{U_i}\to\prod_{i,j} j_*\mathcal{F}|_{U_{i,j}}\to\dots$$

is a resolution of $\mathcal{F}$, which is called **Čech resolution**.

Proof. Proposition 4.17 in [Voi02]. □

Definition 3.3.3. Let $\mathcal{F}$ be a sheaf on X and let $\mathfrak{U}=\{U_i\}_{i\in I}$ be an open cover of X. Then $\mathcal{F}$ is called **acyclic** on $\mathfrak{U}$ if $H^q(U_{i_0\dots i_p},\mathcal{F})=0$ for all $q>0$ and $i_0,\dots,i_p\in I$.

Theorem 3.3.1. Let $\mathcal{F}$ be a sheaf that is acyclic on the open cover $\mathfrak{U}=\{U_i\}_{i\in I}$ of X. Then there is an isomorphism

$$\check{H}^*(\mathfrak{U},\mathcal{F})\cong H^*(X,\mathcal{F}).$$

Proof. Let $\mathcal{I}^\bullet$ be a flabby resolution of $\mathcal{F}$. For each $p\geq 0$, let $0\to\mathcal{I}^p\to\mathcal{I}^{p,\bullet}$ be the Čech resolution, which is also a flabby resolution, since $\mathcal{I}^p$ is flabby and flabby is stable under direct image. Then $H^*(X,\mathcal{F})\cong\mathbb{H}^*(X,\mathcal{I}^\bullet)$ is given by the total cohomology of the double complex $\Gamma(X,\mathcal{I}^{\bullet\bullet})$.

By assumption, the E_1-page of the spectral sequence given by the second filtration is

$$E_1^{p,q}=\begin{cases}C^q(\mathfrak{U},\mathcal{F}), & p=0\\ 0, & p>0.\end{cases}$$

In particular, the spectral sequence degenerates at E_2-page and

$$H^*(X,\mathcal{F})\cong\mathbb{H}^*(X,\mathcal{I}^\bullet)\cong\check{H}^*(\mathfrak{U},\mathcal{F}).$$

□

3.3.3. *Comparison theorem for hypercohomology.*

Definition 3.3.4. Let $\mathcal{F}^\bullet$ be a complex of sheaves and let $\mathfrak{U}=\{U_i\}_{i\in I}$ be an open cover of X. The **Čech hypercohomology** $\check{\mathbb{H}}(\mathfrak{U},\mathcal{F}^\bullet)$ is defined to be the total cohomology of the following double complex:

$$\begin{tikzcd}
 & {} & {} & \\
\cdots \arrow[r] & C^p(\mathfrak{U}, \mathcal{F}^{q+1}) \arrow[r] \arrow[u] & C^{p+1}(\mathfrak{U}, \mathcal{F}^{q+1}) \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & C^p(\mathfrak{U}, \mathcal{F}^{q}) \arrow[r] \arrow[u] & C^{p+1}(\mathfrak{U}, \mathcal{F}^{q}) \arrow[r] \arrow[u] & \cdots \\
 & {} \arrow[u] & {} \arrow[u] &
\end{tikzcd}$$

Theorem 3.3.2. Let $\mathcal{F}^\bullet$ be a complex of sheaves such that each $\mathcal{F}^q$ is acyclic on the open cover $\mathfrak{U} = \{U_i\}_{i \in I}$ of X. Then there is an isomorphism

$$\check{\mathbb{H}}^*(\mathfrak{U}, \mathcal{F}^\bullet) \cong \mathbb{H}^*(X, \mathcal{F}^\bullet).$$

4. Algebraic and analytic geometry

4.1. **Algebraic and analytic variety.**

Definition 4.1.1. A **complex algebraic variety**[3] is a topological space X together with a sheaf $\mathcal{O}_X$ such that

(1) There exists a finite open cover $\{U_i\}_{i\in I}$ of X such that each U_i, together with $\mathcal{O}_X|_{U_i}$, is isomorphic to some affine variety over $\mathbb{C}$ with its sheaf of regular functions.
(2) The diagonal Δ of $X\times X$ is closed in $X\times X$.

Definition 4.1.2. A **complex analytic variety**[4] is a topological space X together with a sheaf $\mathcal{O}_X$ such that

(1) There exists an open cover $\{U_i\}_{i\in I}$ of X such that each U_i, together with $\mathcal{O}_X|_{U_i}$, is isomorphic to some analytic subset[5] in $\mathbb{C}^n$ equipped with its sheaf of holomorphic functions.
(2) The topology of X is Hausdorff.

Definition 4.1.3. Let X be a complex algebraic variety.

(1) A point $x\in X$ is called **non-singular** if $\mathcal{O}_{X,x}$ is a regular local ring.
(2) X is called **smooth** if every point in X is non-singular.

4.2. **Algebraic and analytic coherent sheaves.**

4.2.1. *Definitions.* Let X be a topological space and $\mathcal{A}$ a sheaf of rings on X.

Definition 4.2.1. A sheaf of $\mathcal{A}$-modules $\mathcal{F}$ is said to be **coherent** if

(1) $\mathcal{F}$ is of finite type.
(2) If $s_1,\dots,s_p$ are sections of $\mathcal{F}$ over an open subset $U\subseteq X$, the sheaf of relations between the s_i is of finite type.

Proposition 4.2.1. Every coherent sheaf is locally isomorphic to the cokernel of a homomorphism $\phi\colon \mathcal{A}^q\to\mathcal{A}^p$.

Proof. Proposition 2 in [Ser55], n°12. □

Definition 4.2.2. Let X be a complex algebraic variety. The sheaf of $\mathcal{O}_X$-modules is called an **algebraic sheaf**.

Definition 4.2.3. Let X be a complex analytic variety. The sheaf of $\mathcal{O}_X$-modules is called an **analytic sheaf**.

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[3] in the sense of [Ser55], n°34.

[4] in the sense of [Ser56], n°2.

[5] A subset U of $\mathbb{C}^n$ is analytic if for each $x\in U$, there are functions $f_1,\dots,f_k$, holomorphic in a neighborhood W of x, such that $U\cap W$ is given by the zero locus of $f_i(z)=0$, where $i=1,\dots,k$.

Definition 4.2.4. Let X be a complex algebraic variety. An algebraic sheaf is called **coherent** if it's a coherent sheaf of $\mathcal{O}_X$-modules.

Definition 4.2.5. Let X be a complex analytic variety. An analytic sheaf is called **coherent** if it's a coherent sheaf of $\mathcal{O}_X$-modules.

Theorem 4.2.1 ([Car53])**.** Let X be a Stein manifold[6] and let $\mathcal{F}$ be a coherent analytic sheaf on X. Then

$$H^q(X,\mathcal{F}) = 0$$

for all $q > 0$.

Theorem 4.2.2 ([Ser55])**.** Let X be an affine variety and let $\mathcal{F}$ be a coherent algebraic sheaf on X. Then

$$H^q(X,\mathcal{F}) = 0$$

for all $q > 0$.

4.2.2. *The analytic variety associated to an algebraic variety.* Let $(X,\mathcal{O}_X)$ be a complex algebraic variety equipped with the Zariski topology. Then there is a complex analytic variety structure on X, and X together with this complex analytic structure is denoted by $(X^{an},\mathcal{O}_{X^{an}})$.

Example 4.2.1. If X is a smooth complex algebraic variety, then X^{an} is a complex manifold.

Example 4.2.2. If X is the affine space of dimension n, then $X^{an} = \mathbb{C}^n$.

Example 4.2.3. If X is a smooth affine variety, then X^{an} is a Stein manifold.

Definition 4.2.6. Let $\mathcal{F}$ be an algebraic sheaf on X. The **analytic sheaf associated to** $\mathcal{F}$ is defined by

$$\mathcal{F}^{an} := \epsilon^*\mathcal{F} = \epsilon^{-1}\mathcal{F} \otimes_{\epsilon^{-1}\mathcal{O}_X} \mathcal{O}_{X^{an}},$$

where $\epsilon\colon X^{an} \to X$ is the identity map, which is continuous.

Example 4.2.4. $\mathcal{O}_X^{an} = \mathcal{O}_{X^{an}}$.

Proposition 4.2.2.
(1) The functor $\mathcal{F}^{an}$ is an exact functor.
(2) If $\mathcal{F}$ is a coherent algebraic sheaf, then $\mathcal{F}^{an}$ is a coherent analytic sheaf.

Proof. Proposition 10 in [Ser56], n°9. □

[6] A complex manifold is called Stein if it can be embedded into some $\mathbb{C}^N$.

4.2.3. *Homomorphism induced on cohomology.* Let X be a complex algebraic variety and let $\epsilon\colon X^{an}\to X$ be the identity map. Let $\mathcal{F}$ be an algebraic sheaf on X. If U is a Zariski open subset of X and s is a section of $\mathcal{F}$ over U, one can view s as a section s' of $\epsilon^{-1}\mathcal{F}$ over the open subset U^{an} of X^{an}, and $\alpha(s')=s'\otimes 1$ is a section of $\mathcal{F}^{an}=\epsilon^{-1}\mathcal{F}\otimes\mathcal{O}_{X^{an}}$. The map $s\mapsto\alpha(s')$ is a homomorphism

$$\epsilon\colon\Gamma(U,\mathcal{F})\to\Gamma(U^{an},\mathcal{F}^{an}).$$

Now let $\mathfrak{U}=\{U_i\}_{i\in I}$ be a Zariski open cover of X. Then $\{U_i^{an}\}$ forms an open cover of X^{an}, which we denote by $\mathfrak{U}^{an}$.

For all indices $i_0,\dots,i_q\in I$, there is a canonical homomorphism

$$\epsilon\colon\Gamma(U_{i_0}\cap\dots U_{i_q},\mathcal{F})\to\Gamma(U_{i_0}^{an}\cap\dots U_{i_q}^{an},\mathcal{F}^{an}),$$

and hence a homomorphism

$$\epsilon\colon C^*(\mathfrak{U},\mathcal{F})\to C^*(\mathfrak{U}^{an},\mathcal{F}^{an}).$$

This homomorphism commutes with the coboundary δ, and so defines, by passing to cohomology, the homomorphisms

$$\epsilon\colon H^*(\mathfrak{U},\mathcal{F})\to H^*(\mathfrak{U}^{an},\mathcal{F}^{an}).$$

Finally, by passing to the direct limit over $\mathfrak{U}$, one obtains the homomorphisms induced on cohomology groups

$$\epsilon\colon H^*(X,\mathcal{F})\to H^*(X^{an},\mathcal{F}^{an}).\tag{4.1}$$

Remark 4.2.1. These homomorphisms also induce homomorphisms between hypercohomology groups of complexes of sheaves.

4.2.4. *GAGA principle.*

Theorem 4.2.3 ([Ser56])**.** Let X be a projective variety. For any coherent algebraic sheaf on X, the homomorphism in (4.1)

$$\epsilon\colon H^*(X,\mathcal{F})\to H^*(X^{an},\mathcal{F}^{an})$$

is an isomorphism.

Theorem 4.2.4 ([Ser56])**.** Let X be a projective variety. If $\mathcal{F}$ and $\mathcal{G}$ are two coherent algebraic sheaves on X, every analytic homomorphism of $\mathcal{F}^{an}$ to $\mathcal{G}^{an}$ comes from a unique algebraic homomorphism of $\mathcal{F}$ to $\mathcal{G}$.

Theorem 4.2.5 ([Ser56])**.** Let X be a projective variety. For every coherent analytic sheaf $\mathcal{G}$ on X^{an}, there exists a coherent algebraic sheaf on X such that $\mathcal{F}^{an}$ is isomorphic to $\mathcal{G}$. Moreover, this property determines $\mathcal{F}$ up to isomorphisms.

4.3. **Algebraic de Rham complex.**

4.3.1. *Kähler differentials.* Let R be a $\mathbb{C}$-algebra.

Definition 4.3.1. The **module of Kähler differentials** of R over $\mathbb{C}$ is an R-module $\Omega^1_{R/\mathbb{C}}$, together with a derivation $\mathrm{d}\colon R \to \Omega^1_{R/\mathbb{C}}$ satisfying the following universal property: for any R-module M and any derivation $\mathrm{d}'\colon R \to M$, there exists a unique R-module morphism $f\colon \Omega^1_{R/\mathbb{C}} \to M$ such that $\mathrm{d}' = f \circ \mathrm{d}$.

Remark 4.3.1. The module of Kähler differentials of R over $\mathbb{C}$ can be constructed as the free R-module generated by the symbols $\{\mathrm{d}f \mid f \in R\}$, modulo the submodule generated by all expressions of the form

(1) $\mathrm{d}c$ for all $c \in \mathbb{C}$.
(2) $\mathrm{d}(cf+g) = c\mathrm{d}f + \mathrm{d}g$ for all $c \in \mathbb{C}, f, g \in R$.
(3) $\mathrm{d}(fg) = f\mathrm{d}g + g\mathrm{d}f$ for all $f, g \in R$.

Example 4.3.1. If $R = \mathbb{C}[x_1, \ldots, x_n]$, then

$$\Omega^1_{R/\mathbb{C}} \cong \bigoplus_{i=1}^{n} R\,\mathrm{d}x_i.$$

Proposition 4.3.1 ([Mat86, Theorem 25.2])**.** Let $R \to S$ be a surjective morphism of $\mathbb{C}$-algebras with kernel I. Then there is an exact sequence of S-modules

$$I/I^2 \to S \otimes_R \Omega^1_{R/\mathbb{C}} \to \Omega^1_{S/\mathbb{C}} \to 0,$$

where $[f] \in I/I^2$ maps to $1 \otimes \mathrm{d}f \in S \otimes_R \Omega^1_{R/\mathbb{C}}$.

Corollary 4.3.1. Let $R = \mathbb{C}[x_1, \ldots, x_n]$ and $S = R/I$, where $I = \langle f_1, \ldots, f_s \rangle$. Then there is an exact sequence

$$I/I^2 \to S^n \to \Omega^1_{S/\mathbb{C}} \to 0$$

4.3.2. *Sheaf of Kähler differentials.* Let X be a complex algebraic variety of dimension n.

Definition 4.3.2. The **cotangent sheaf** Ω^1_X is the sheaf of $\mathcal{O}_X$-modules defined by

$$\Omega^1_X(U) := \Omega^1_{\mathcal{O}_X(U)/\mathbb{C}}$$

on affine open subsets U.

Definition 4.3.3. The **tangent sheaf** $\mathcal{T}_X$ is defined by $\mathcal{T}_X = \mathscr{H}om(\Omega^1_X, \mathcal{O}_X)$.

Definition 4.3.4. The **sheaf of k-forms** Ω^k_X is the sheaf of $\mathcal{O}_X$-modules defined by the wedge product $\bigwedge^k \Omega^1_X$.

Definition 4.3.5. The derivation $\mathrm{d}\colon \mathcal{O}_X \to \Omega^1_X$ extends to a complex

$$\mathcal{O}_X \xrightarrow{\mathrm{d}} \Omega^1_X \xrightarrow{\mathrm{d}} \Omega^2_X \xrightarrow{\mathrm{d}} \ldots \xrightarrow{\mathrm{d}} \Omega^n_X,$$

which is called the **algebraic de Rham complex**.

Theorem 4.3.1 ([Har75])**.** Let X be the affine space of dimension n. Then

$$0 \to \mathbb{C} \to \Gamma(X, \mathcal{O}_X) \to \Gamma(X, \Omega_X^1) \to \cdots \to \Gamma(X, \Omega_X^n) \to 0$$

is an exact sequence of $\mathbb{C}$-vector spaces.

Theorem 4.3.2 ([Har77, Theorem 8.15 in ChapterII])**.** The algebraic variety X is smooth if and only if Ω_X^1 is locally free of rank n.

5. Local system and integrable connection

In this section we always assume X is a complex manifold and $\mathcal{A}_X^p$ is the locally free sheaf of smooth p-forms.

Definition 5.1. A sheaf $\mathcal{V}$ on X is called a locally constant sheaf of rank r valued in $\mathbb{C}$, if for each point $x \in X$, there is an open subset U containing x such that $\mathcal{V}|_U$ is constant sheaf $\underline{\mathbb{C}^r}$.

Remark 5.1. In other words, there exists an open cover $\{U_\alpha\}$ such that $\mathcal{V}|_{U_\alpha}$ is isomorphic to the constant sheaf $\underline{\mathbb{C}^r}$. Then the local system $\mathcal{V}$ is completely determined by the transition functions $g_{\alpha\beta}\colon U_{\alpha\beta} \to \mathrm{GL}_n(\mathbb{C})$, which are locally constant functions.

Definition 5.2. Let $\mathcal{E}$ be a locally free sheaf on X. A **connection** is a $\mathbb{C}$-linear map

$$\nabla\colon \mathcal{E} \to \mathcal{A}_X^1 \otimes \mathcal{E}$$

satisfying the following condition

$$\nabla(\varphi \otimes e) = \mathrm{d}\varphi \otimes e + \varphi \nabla e$$

for all sections e of $\mathcal{E}$ and φ of $\mathcal{O}_X$.

Remark 5.2. The definition of ∇ extends to $\nabla\colon \mathcal{A}_X^p \otimes \mathcal{E} \to \mathcal{A}_X^{p+1} \otimes \mathcal{E}$ by defining

$$\nabla(\omega \otimes e) = \mathrm{d}\omega \otimes e + (-1)^p \omega \wedge \nabla e$$

for all sections ω of $\mathcal{A}_X^p$ and sections e of $\mathcal{E}$.

Remark 5.3. Let $\{e_\alpha\}$ be a local frame of $\mathcal{E}$. For any section $s = s^\alpha e_\alpha$ of $\mathcal{E}$, one has

$$\nabla(s^\alpha e_\alpha) = \mathrm{d}s^\alpha e_\alpha + s^\alpha \nabla e_\alpha.$$

Thus the connection ∇ is completely determined by

$$\nabla e_\alpha = \omega_\alpha^\beta e_\beta,$$

where ω_α^β are 1-forms, which form a (smooth) matrix-valued 1-form ω.

Definition 5.3. A connection ∇ is **integrable** if its curvature $\nabla^2\colon \mathcal{E} \to \mathcal{A}_X^2 \otimes \mathcal{E}$ vanishes.

Remark 5.4. Let $\{e_\alpha\}$ be a local frame of $\mathcal{E}$. For any section $s = s^\alpha e_\alpha$ of $\mathcal{E}$, one has

$$\begin{aligned}
\nabla^2(s^\alpha e_\alpha) &= \nabla(\mathrm{d}s^\alpha \otimes e_\alpha + s^\alpha \omega_\alpha^\beta \otimes e_\beta) \\
&= -\mathrm{d}s^\alpha \wedge \omega_\alpha^\beta \otimes e_\beta + \mathrm{d}(s^\alpha \omega_\alpha^\beta) \otimes e_\beta - s^\alpha \omega_\alpha^\beta \wedge \omega_\beta^\gamma \otimes e_\gamma \\
&= s^\alpha(\mathrm{d}\omega_\alpha^\beta - \omega_\alpha^\gamma \wedge \omega_\gamma^\beta) \otimes e_\beta, \\
\nabla^2(e_\alpha) &= \nabla(\omega_\alpha^\beta \otimes e_\beta) \\
&= \mathrm{d}\omega_\alpha^\beta \otimes e_\beta - \omega_\alpha^\beta \wedge \nabla e_\beta \\
&= \mathrm{d}\omega_\alpha^\beta \otimes e_\beta - \omega_\alpha^\beta \wedge \omega_\beta^\gamma \otimes e_\gamma \\
&= (\mathrm{d}\omega_\alpha^\beta - \omega_\alpha^\gamma \wedge \omega_\gamma^\beta) \otimes e_\beta.
\end{aligned}$$

This shows ∇^2 is a global section of $\mathcal{A}_X^2 \otimes \underline{\mathrm{End}}_{\mathcal{O}_X}(\mathcal{E})$, which is locally given by $\mathrm{d}\omega - \omega \wedge \omega$.

Definition 5.4. A locally free sheaf together with an integrable connection is called a **flat bundle**.

Proposition 5.1. Let ∇ be an integrable connection on a locally free sheaf $\mathcal{E}$ on X. Then the sheaf of horizontal sections $\mathcal{E}^{\nabla=0}$ is a local system.

Proposition 5.2. Let $\mathcal{V}$ be a local system on X. Then the locally free sheaf $\mathcal{E} := \mathcal{O}_X \otimes \mathcal{V}$, together with the canonical connection $\nabla_{\mathrm{can}}(f \otimes s) := \mathrm{d}f \otimes s$, is a flat bundle.

Theorem 5.1. The functor $(\mathcal{E}, \nabla) \mapsto \mathcal{E}^{\nabla=0}$ is an equivalence between the category of flat bundles and the category of complex local systems, with quasi-inverse $\mathcal{V} \mapsto (\mathcal{O}_X \otimes \mathcal{V}, \nabla_{\mathrm{can}})$.

Proposition 5.3. Let $\mathcal{V}$ be a local system on X. Then

$$H^*(X, \mathcal{V}) \cong \mathbb{H}^*(X, \mathcal{A}_X^\bullet \otimes \mathcal{V}).$$

Proof. Note the following complex of sheaves

$$0 \to \mathcal{V} \to \mathcal{A}_X^\bullet \otimes \mathcal{V}$$

gives a resolution of $\mathcal{V}$ by coherent sheaves. $\square$

Part 2. Comparison theorems

6. Grothendieck's version

6.1. **Introduction.** In this section, we will prove the following comparison theorem.

Theorem 6.1.1 ([Gro66a]). Let X be a smooth complex variety and X^{an} be the underlying complex manifold. Then there is the following isomorphism

$$H^*(X^{an},\mathbb{C}) \cong \mathbb{H}^*(X,\Omega_X^\bullet),$$

where $\Omega_X^\bullet$ is the algebraic de Rham complex.

The following is a sketch of the proof. First, we use Čech cohomology arguments to reduce to the case where X is a smooth affine variety, and then we embed X into some smooth projective variety Y by $j\colon X \to Y$. For smooth projective varieties, the GAGA principle (Theorem 4.2.3) closely relates algebraic geometry and analytic geometry.

To be precise, let X be a smooth affine variety. Then its projectivization $\overline{X}$ is a projective variety, but it may have singularities. By Hironaka's resolution of singularities ([Hir64]), there is a surjective regular map $\pi\colon Y \to \overline{X}$, where Y is a smooth projective variety such that $\pi^{-1}(\overline{X}-X)$ is a simple normal crossing divisor[7] $D \subset Y$, and the restriction $\pi|_{Y-D}$ is an isomorphism.

The proof of the affine case is divided into the following three steps.

(1) First, we establish the following two isomorphisms.

Theorem 6.1.2. There is an isomorphism between hypercohomology groups

$$\mathbb{H}^*(Y, j_*\Omega_X^\bullet) \cong \mathbb{H}^*(X,\Omega_X^\bullet).$$

Proof. Let $\mathfrak{V} = \{V_i\}_{i\in I}$ be an affine open cover of Y. Since j is an affine morphism, the direct image $j_*\Omega_X^\bullet$ is a complex of coherent sheaves on Y, and thus by Serre's theorem (Theorem 4.2.2) and Theorem 3.3.2, one has

$$\mathbb{H}^*(Y, j_*\Omega_X^\bullet) = \check{H}^*(\mathfrak{V}, j_*\Omega_X^\bullet).$$

On the other hand, $\mathfrak{U} = \{V_i \cap X\}_{i\in I}$ is also an affine open cover of X, since X is affine and the intersection of affine varieties is still affine. By the same argument one has

$$\mathbb{H}^*(X,\Omega_X^\bullet) = \check{H}(\mathfrak{U},\Omega_X^\bullet).$$

[7] A divisor D is called a **simple normal crossing divisor** if locally there exist coordinates $\{z_1,\dots,z_n\}$ on X such that D is defined by the equation $z_1 \dots z_r = 0$, for an integer r that may depend on the chosen open set.

This completes the proof, since by definition one has

$$\check{H}^*(\mathfrak{V}, j_*\Omega_X^\bullet) = \check{H}(\mathfrak{U}, \Omega_X^\bullet).$$

□

Theorem 6.1.3.

$$\mathbb{H}^*(Y^{an}, j_*^{an}\Omega_{X^{an}}^\bullet) \cong H^*(X^{an}, \mathbb{C}).$$

Proof. Since Cartan's theorem (4.2.1) is the analytic analogue of Serre's theorem, the same argument as above proves

$$\mathbb{H}^*(Y^{an}, j_*^{an}\Omega_{X^{an}}^\bullet) \cong \mathbb{H}^*(X^{an}, \Omega_{X^{an}}^\bullet).$$

On the other hand, by the holomorphic Poincaré lemma, $\Omega_{X^{an}}^\bullet$ gives a resolution of the constant sheaf $\underline{\mathbb{C}}$, and thus $\mathbb{H}^*(X^{an}, \Omega_{X^{an}}^\bullet) \cong H^*(X^{an}, \mathbb{C})$. □

(2) Second, we use direct-limit arguments and the GAGA principle to prove the following isomorphism.

Theorem 6.1.4.

$$\mathbb{H}^*(Y^{an}, j_*^m\Omega_{X^{an}}^\bullet) \cong \mathbb{H}^*(Y, j_*\Omega_X^\bullet),$$

where $j_*^m\Omega_{X^{an}}^\bullet := (j_*\Omega_X^\bullet)^{an}$ is exactly the sheaf of sections of $\Omega_{X^{an}}^\bullet$ that are meromorphic along D.

Proof. For each p, one has $j_*\Omega_X^p = \varinjlim_n \Omega_Y^p(nD)$, where $\Omega_Y^p(nD)$ is the sheaf of p-forms on Y that are regular on X and admit poles along D of order $\leq n$. Similarly, one has $j_*^m\Omega_{X^{an}}^p = \varinjlim_n \Omega_{Y^{an}}^p(nD)$.

On the other hand, direct limit commutes with cohomology (Proposition 2.9 in ChapterIII of [Har77]). Then for any q, one has

$$\begin{aligned} H^q(Y^{an}, j_*^m\Omega_{X^{an}}^p) &\cong H^q(Y^{an}, \varinjlim_n \Omega_{Y^{an}}^p(nD)) \\ &\cong H^q(Y, \varinjlim_n \Omega_Y^p(nD)) \\ &\cong H^q(Y, j_*\Omega_X^p). \end{aligned}$$

This gives an isomorphism between the E_1-pages of the spectral sequences, compatible with the differentials. Hence it gives an isomorphism between the E_∞-pages (Proposition 2.3), that is,

$$\mathbb{H}^*(Y^{an}, j_*^m\Omega_{X^{an}}^\bullet) \cong \mathbb{H}^*(Y, j_*\Omega_X^\bullet).$$

□

(3) The key step is to prove the following isomorphism

Theorem 6.1.5.

$$\mathbb{H}^*(Y^{an}, j_*^m \Omega^\bullet_{X^{an}}) \cong \mathbb{H}^*(Y^{an}, j_*^{an} \Omega^\bullet_{X^{an}}).$$

In summary, we prove the following diagram.

$$\begin{tikzcd}
H^*(Y, j_*\Omega^\bullet_X) \arrow[d, "6.1.4"'] & & \mathbb{H}^*(X, \Omega^\bullet_X) \arrow[ll, "6.1.2"'] \arrow[dd, dashed, no head] \\
\mathbb{H}^*(Y^{an}, j_*^m \Omega^\bullet_{X^{an}}) \arrow[d, "6.3.2"'] & & \\
\mathbb{H}^*(Y^{an}, j_*^{an} \Omega^\bullet_{X^{an}}) \arrow[rr, "6.1.3"] & & H^*(X^{an}, \mathbb{C}).
\end{tikzcd}$$

6.2. **Reduce to affine case.** Let X be a smooth complex variety and let $\mathfrak{U} = \{U_i\}_{i\in I}$ be an affine open cover of X. Since coherent sheaves are acyclic on affine pieces (Theorem 4.2.2), the algebraic de Rham complex $\Omega^\bullet_X$ is acyclic with respect to $\mathfrak{U}$, and thus by Theorem 3.3.2 one has

$$\check{\mathbb{H}}^*(\mathfrak{U}, \Omega^\bullet_X) \cong \mathbb{H}^*(X, \Omega^\bullet_X).$$

Similarly, since coherent sheaves are acyclic on Stein manifolds, there is the following isomorphism

$$\check{\mathbb{H}}^*(\mathfrak{U}^{an}, \Omega^\bullet_{X^{an}}) \cong \mathbb{H}^*(X^{an}, \Omega^\bullet_{X^{an}}).$$

Note that the E_1-page of spectral sequences converging to $\check{\mathbb{H}}^*(\mathfrak{U}, \Omega^\bullet_X)$ is

$$\begin{aligned}
E_1^{p,q} &= H^q_{\mathrm{d}}\left(C^p(\mathfrak{U}, \Omega^\bullet_X)\right) \\
&= \prod_{i_0,\dots,i_p} H^q_{\mathrm{d}}(\Gamma(U_{i_0\dots i_p}, \Omega^\bullet_X)) \\
&= \prod_{i_0,\dots,i_p} \mathbb{H}^q(\Gamma(U_{i_0\dots i_p}, \Omega^\bullet_X)),
\end{aligned}$$

where the last equality holds since $U_{i_0\dots i_p}$ is affine, and thus $\Omega^\bullet_X$ is acyclic on $U_{i_0\dots i_p}$. Similarly the E_1-page of spectral sequences converging to $\check{\mathbb{H}}^*(\mathfrak{U}^{an}, \Omega^\bullet_{X^{an}})$ is

$$\begin{aligned}
(E'_1)^{p,q} &= H^q_{\mathrm{d}}\left(C^p(\mathfrak{U}^{an}, \Omega^\bullet_{X^{an}})\right) \\
&= \prod_{i_0,\dots,i_p} H^q_{\mathrm{d}}(\Gamma(U^{an}_{i_0\dots i_p}, \Omega^\bullet_{X^{an}})) \\
&= \prod_{i_0,\dots,i_p} H^q(U^{an}_{i_0\dots i_p}, \mathbb{C}).
\end{aligned}$$

The proof of the affine case shows the following map

$$\epsilon\colon \mathbb{H}^*(U_{i_0\dots i_p}, \Omega^\bullet_X) \to \mathbb{H}^*(U^{an}_{i_0\dots i_p}, \Omega^\bullet_{X^{an}}) \cong H^*(U^{an}_{i_0\dots i_p}, \mathbb{C})$$

is an isomorphism, and it also commutes with differentials in spectral sequences. By Proposition 2.3, it induces an isomorphism between the E_∞-pages, that is, an isomorphism

$$\mathbb{H}^*(X, \Omega_X^\bullet) \cong H^*(X^{an}, \mathbb{C}).$$

6.3. **Logarithmic differential forms.**

6.3.1. *Logarithmic differential forms.* To prove the key step, we first introduce differential forms with logarithmic poles along a simple normal crossing divisor. Let Y be a complex manifold with a simple normal crossing divisor D and let $j\colon X = Y - D \to Y$.

Definition 6.3.1. The sheaf of **differential k-forms with logarithmic poles along** D, denoted by $\Omega_Y^k(\log D)$, is the subsheaf of $j_*^m \Omega_X^k$ defined by the following condition:

- If $\alpha \in \Gamma(U, j^m \Omega_X^k)$ for some open subset $U \subseteq Y$, then $\alpha \in \Gamma(U, \Omega_Y^k(\log D))$ if and only if α admits a pole of order at most 1 along D, and the same holds for $\mathrm{d}\alpha$.

Lemma 6.3.1. Let $\{z_1, \dots, z_n\}$ be a local coordinate on an open subset $U \subseteq Y$, in which $D \cap U$ is defined by the equation $z_1 \dots z_r = 0$. For convenience we denote

$$\delta_j = \begin{cases} \mathrm{d}z_j/z_j, & j \le r \\ \mathrm{d}z_j, & j > r, \end{cases}$$

and for $I = \{j_1, \dots, j_k\} \subseteq \{1, \dots, n\}$ with $j_1 < \dots < j_s$, we denote

$$\delta_I = \delta_{j_1} \wedge \dots \wedge \delta_{j_k}.$$

Then $\Omega_Y^k(\log D)|_U$ is a sheaf of free $\mathcal{O}_U$-modules with basis $\{\delta_I\}_{|I|=k}$

Proof. Lemma 8.16 in [Voi02]. □

Corollary 6.3.1. The sheaves $\Omega_Y^k(\log D)$ are sheaves of locally free $\mathcal{O}_Y$-modules.

Proof. Corollary 8.17 in [Voi02]. □

6.3.2. *The proof of the key step.* Note that there is a natural inclusion

$$\Omega_Y^\bullet(\log D) \subseteq j_* \Omega_X^\bullet, \tag{6.1}$$

which is compatible with differentials.

Theorem 6.3.1. The morphism (6.1) is a quasi-isomorphism.

Proof. To show $\mathcal{H}^k(\Omega_Y^\bullet(\log D)) \cong \mathcal{H}^k(j_*\Omega_X^\bullet)$ for each k, it suffices to check this stalk by stalk. If $x \in X$, it is easy to show that

$$(\mathcal{H}^k(\Omega_Y^\bullet(\log D)))_x \cong (\mathcal{H}^k(j_*\Omega_X^\bullet))_x \cong \mathbb{C}$$

unless $k = 0$ by holomorphic Poincaré lemma.

Now suppose $x \in D$. Without loss of generality, we may assume that $Y = D_1 \times \cdots \times D_n$ is a polydisk of dimension n and $D = \{(z_1, \ldots, z_n) \mid z_1 \ldots z_r = 0\}$. Thus $X = Y - D = D_1^* \times \cdots \times D_r^* \times D_{r+1} \times \cdots \times D_n$ retracts onto the product of circles $T_r = (S^1)^r = \prod_{i \le r} \partial D_i$, and the cohomology of this product of circles is given by

$$H^1(T_r, \mathbb{C}) \cong \mathbb{C}^r$$
$$H^k(T_r, \mathbb{C}) \cong \bigwedge^k H^1(T_r, \mathbb{C}),$$

where the second equality can be obtained by Künneth formula easily.

First, let us prove that the morphism (6.1) induces a surjective map in cohomology. For the closed 1-forms $\omega_i = \mathrm{d}z_i/z_i \in \Gamma(Y, \Omega_Y^1(\log D))$, their integrals over the circles ∂D_i satisfy

$$\int_{\partial D_j} \omega_i = 2\pi\sqrt{-1}\delta_{ij},$$

and thus the classes of these forms generate $H^1(T_r, \mathbb{C})$. The exterior products $\omega_I = \wedge_{i \in I} \omega_i \in \Gamma(Y, \Omega_Y^k(\log D))$ also generate the cohomology of T_r in degree k, since $H^k(T_r, \mathbb{C}) = \bigwedge^k H^1(T_r, \mathbb{C})$. This proves surjectivity.

For injectivity, we need to show that if a closed form $\alpha \in \Gamma(Y, \Omega_Y^k(\log D))$ is d-exact in $\Gamma(Y, j_*\Omega_X^\bullet) = \Gamma(X, \Omega_X^\bullet)$, then it is d-exact in $\Gamma(Y, \Omega_Y^k(\log D))$. The proof of injectivity is by induction on r.

If $r = 0$, the statement holds from the holomorphic Poincaré lemma. Now assume the statement holds for $r - 1$. For any $\alpha \in \Gamma(Y, \Omega_Y^k(\log D))$, we decompose it into

$$\alpha = \frac{\mathrm{d}z_r}{z_r} \wedge \beta + \gamma,$$

where $\mathrm{d}z_r$ does not occur in β, and the coefficients of β are independent of z_r, while γ is holomorphic in z_r.

(1) If α is closed, one has

$$\mathrm{d}\alpha = \frac{\mathrm{d}z_r}{z_r} \wedge \mathrm{d}\beta + \mathrm{d}\gamma = 0.$$

Note that γ is holomorphic in z_r and the coefficients of β are independent of z_r, so it leads to $\mathrm{d}\beta = 0$ and $\mathrm{d}\gamma = 0$.

(2) If α is d-exact in $\Gamma(X,\Omega_X^\bullet)$, say $\alpha = \mathrm{d}\widetilde{\alpha}$, then we decompose as

$$\alpha = \mathrm{d}\widetilde{\alpha} = \mathrm{d}\left(\frac{\mathrm{d}z_r}{z_r}\wedge\widetilde{\beta}+\widetilde{\gamma}\right) = \frac{\mathrm{d}z_r}{z_r}\wedge \mathrm{d}\widetilde{\beta}+\mathrm{d}\widetilde{\gamma}.$$

For the same reason, both β and γ are d-exact in $\Gamma(X,\Omega_X^\bullet)$.

Now suppose $\alpha \in \Gamma(Y,\Omega_Y^k(\log D))$ is closed and write $\alpha = \mathrm{d}z_r/z_r\wedge\beta+\gamma$. Since γ is closed and holomorphic in z_r, it may be regarded as a closed holomorphic k-form with logarithmic poles along $D' = \{z \mid z_1\dots z_{r-1} = 0\}$, and thus by the induction hypothesis it is d-exact in $\Gamma(Y,\Omega_Y^k(\log D))$. By the same argument, one can show that $\mathrm{d}z_r/z_r\wedge\beta$ is also d-exact in $\Gamma(Y,\Omega_Y^k(\log D))$. This completes the proof. □

On the other hand, there is also a natural inclusion

$$\Omega_Y^\bullet(\log D)\subseteq j_*^m\Omega_X^\bullet, \tag{6.2}$$

which is compatible with differentials.

Theorem 6.3.2. The morphism (6.2) is a quasi-isomorphism.

Proof. For the same reason, we may assume X and Y are as before. As seen above, the stalk of $\mathcal{H}^1(\Omega_Y^\bullet(\log D))$ is generated by $[\mathrm{d}z_1/z_1],\dots,[\mathrm{d}z_r/z_r]$. We now prove that the morphism (6.2) is a quasi-isomorphism by induction on r; the case $r=0$ is clear.

Lemma 6.3.2. Let $\varphi\in\Gamma(Y,j_*^m\Omega_X^k)$ be a closed meromorphic k-form on Y that is holomorphic on X. Then there exist closed meromorphic forms $\varphi_0\in\Gamma(Y,j_*^m\Omega_X^k)$ and $\alpha_1\in\Gamma(Y,j_*^m\Omega_X^{k-1})$, which have no poles along z_r, such that

$$[\varphi] = [\varphi_0] + \left[\frac{\mathrm{d}z_r}{z_r}\right]\wedge[\alpha_1].$$

Proof. First we write

$$\varphi = \mathrm{d}z_r\wedge\alpha+\beta,$$

where α is a meromorphic $(k-1)$-form and β does not involve $\mathrm{d}z_r$. Then expand α and β as Laurent series in z_r:

$$\begin{aligned}\alpha &= \alpha_0+\alpha_1 z_r^{-1}+\cdots+\alpha_\ell z_r^{-\ell}\\ \beta &= \beta_0+\beta_1 z_r^{-1}+\cdots+\beta_\ell z_r^{-\ell},\end{aligned}$$

where α_i and β_i for $1\le i\le\ell$ do not involve z_r or dz_r and are meromorphic in the other variables, and α_0,β_0 are holomorphic in z_r, are meromorphic in the other variables, and do not involve $\mathrm{d}z_r$. Thus

$$\varphi = \varphi_0 + \left(\mathrm{d}z_r\wedge\sum_{i=1}^r\alpha_i z_r^{-i}+\sum_{i=1}^r\beta_i z_r^{-i}\right),$$

where $\varphi_0 = \mathrm{d}z_r \wedge \alpha_0 + \beta_0$. By comparing the coefficients of $z_r^{-i}\mathrm{d}z_r$ and z_r^{-i}, one can deduce from $\mathrm{d}\varphi$,

$$\mathrm{d}\alpha_1 = \mathrm{d}\alpha_2 + \beta_1 = \mathrm{d}\alpha_3 + 2\beta_2 = \cdots = r\beta_r = 0$$
$$\mathrm{d}\beta_1 = \mathrm{d}\beta_2 = \cdots = \mathrm{d}\beta = 0,$$

and $\mathrm{d}\varphi_0 = 0$. If we write

$$\varphi = \varphi_0 + \frac{\mathrm{d}z_r}{z_r} \wedge \alpha_1 + \left(\mathrm{d}z_r \wedge \sum_{i=2}^{r} \alpha_i z_r^{-i} + \sum_{i=1}^{r} \beta_i z_r^{-i}\right)$$

It turns out that

$$\theta = -\frac{\alpha_2}{z_r} - \frac{\alpha_3}{2z_r^2} - \cdots - \frac{\alpha_r}{(r-1)z_r^{r-1}}$$

satisfies

$$\varphi - \varphi_0 - \frac{\mathrm{d}z_r}{z_r} \wedge \alpha_1 = \mathrm{d}\theta.$$

□

Since φ_0 and α_1 are meromorphic forms with no poles along $z_r = 0$, their singularity set is contained in the simple normal crossing divisor $z_1 \ldots z_{r-1} = 0$. By induction on r, the cohomology classes of φ_0 and φ_1 are generated by $[\mathrm{d}z_1/z_1], \ldots, [\mathrm{d}z_{r-1}/z_{r-1}]$. This completes the proof. □

Corollary 6.3.2.

$$\mathbb{H}^*(Y^{an}, j_*^m \Omega^\bullet_{X^{an}}) \cong \mathbb{H}^*(Y^{an}, j_*^{an} \Omega^\bullet_{X^{an}}).$$

Remark 6.3.1. There is also an analogue in the algebraic setting.

Theorem 6.3.3. Let X be a smooth complex algebraic variety and $j\colon X \to Y$ be an inclusion such that $D = Y - X$ is simple normal crossing. Then

$$\mathbb{H}^*(Y, \Omega^\bullet_Y(D)) \cong \mathbb{H}^*(Y, j_*\Omega^\bullet_X).$$

7. Local system valued version

7.1. **Introduction.** In this section we will prove the local system valued version of Grothendieck's comparison theorem, and the ideas are almost the same as before.

Theorem 7.1.1 ([Del70])**.** Let X be a smooth complex algebraic variety and X^{an} be the corresponding complex manifold. Let $\mathcal{E}$ be a locally free sheaf on X equipped with a regular integrable connection ∇ and $\mathcal{V} := (\mathcal{E}^{an})^{\nabla^{an}=0}$ be the local system of horizontal sections on the underlying complex manifold X^{an}. Then there is the following isomorphism

$$H^*(X^{an}, \mathcal{V}) \cong \mathbb{H}^*(X, \Omega_X^\bullet(\mathcal{E})).$$

The idea of the proof is the same as before. First, by the same argument, we can reduce to the case where X is smooth affine with an embedding $j\colon X \to Y$, where Y is a smooth projective variety and $D = Y - X$. Then we need to go through the following diagram:

$$\begin{tikzcd}
H^*(Y, j_*\Omega_X^\bullet(\mathcal{E})) \arrow[d, "2"'] & & \mathbb{H}^*(X, \Omega_X^\bullet(\mathcal{E})) \arrow[ll, "1"'] \arrow[dd, dashed] \\
\mathbb{H}^*(Y^{an}, j_*^m\Omega_{X^{an}}^\bullet(\mathcal{E}^{an})) \arrow[d, "3"'] & & \\
\mathbb{H}^*(Y^{an}, j_*^{an}\Omega_{X^{an}}^\bullet(\mathcal{E}^{an})) \arrow[rr, "4"] & & H^*(X^{an}, \mathcal{V}).
\end{tikzcd}$$

The proof of (1), (2) and (4) is the same as before, and the most difficult step is to show the morphism of complexes

$$j_*^m\Omega_{X^{an}}^\bullet(\mathcal{E}) \to j_*\Omega_{X^{an}}^\bullet(\mathcal{E})$$

is a quasi-isomorphism. In this section, we will define the so-called **Deligne's canonical extension** $\widetilde{\mathcal{E}}$ of $\mathcal{E}$ on Y^{an}, and prove

$$\Omega_{Y^{an}}^\bullet(\log D)(\widetilde{\mathcal{E}}) \to j_*^m\Omega_{X^{an}}^\bullet(\mathcal{E}) \to j_*^{an}\Omega_{X^{an}}^\bullet(\mathcal{E})$$

are quasi-isomorphisms.

7.2. **Local system valued logarithmic pole differential.** For simplicity, we write X, Y instead of X^{an}, Y^{an}, since we focus on the analytic setting, and the simple normal crossing divisor $D = Y - X$ is written as $D_1 + \cdots + D_r$.

7.2.1. *Residue map.*

Definition 7.2.1. For every $k \geq 1$, the **residue map along** D_i is a map

$$\mathrm{Res}_i\colon \Omega_Y^k(\log D) \to \Omega_{D_i}^{k-1}(\log(D - D_i)|_{D_i})$$

such that if φ is a local section of $\Omega^k_Y(\log D)$, we write it as

$$\varphi = \varphi_1 + \varphi_2 \wedge \frac{\mathrm{d}z_i}{z_i},$$

where φ_1 lies in the span of the δ_I with $i \notin I$ and $\varphi_2 = \sum_{i \in I} a_I \delta_{I-\{i\}}$, then

$$\mathrm{Res}_{D_i}(\varphi) = \sum a_I \delta_{I-\{1\}}|_{D_i}.$$

Remark 7.2.1.

(1) Let $\mathcal{E}$ be a locally free sheaf on Y. Then the residue map extends to

$$\mathrm{Res}_i : \Omega^k_Y(\log D)(\mathcal{E}) \to \Omega^{k-1}_{D_i}(\log(D - D_i)|_{D_i})(\mathcal{E})$$

linearly.

(2) In particular, if $k = 1$, the residue map along D_i is a map

$$\mathrm{Res}_i : \Omega^1_Y(\log D)(\widetilde{\mathcal{E}}) \to \mathcal{O}_{D_i} \otimes \widetilde{\mathcal{E}}.$$

7.2.2. *Residue of the connection with logarithmic poles.* Let $\mathcal{E}$ be a locally free sheaf on X and let ∇ be an integrable connection on $\mathcal{E}$. Suppose that $\mathcal{E}$ is given as the restriction of a locally free sheaf $\widetilde{\mathcal{E}}$ on Y. Locally on X, a choice of basis e of $\widetilde{\mathcal{E}}$ gives a connection matrix $\Gamma \in j_*\Omega^1_X(\underline{\mathrm{End}}(\widetilde{\mathcal{E}}))$, and a change of basis $e \mapsto e'$ modifies Γ by a section of $\Omega^1_Y(\underline{\mathrm{End}}(\widetilde{\mathcal{E}}))$. Thus the "pole part" of Γ depends only on $\widetilde{\mathcal{E}}$ and ∇.

Definition 7.2.2. The connection ∇ has **at worst logarithmic poles along** D if the connection forms present at worst logarithmic poles along D.

Definition 7.2.3. The residue of the connection Γ along a local component D_i of D is defined by

$$\mathrm{Res}_i(\Gamma) \in \underline{\mathrm{End}}(\widetilde{\mathcal{E}}|_{D_i}),$$

which depends only on $\widetilde{\mathcal{E}}$ and ∇.

Theorem 7.2.1. Let $\widetilde{\mathcal{E}}$ be a locally free sheaf on Y and let ∇ be an integrable connection on $\widetilde{\mathcal{E}}|_X$ that has at worst logarithmic poles along D. If the residues of the connection form Γ along all components of D do not admit any strictly positive integer as an eigenvalue, then

$$\Omega^\bullet_Y(\log D)(\widetilde{\mathcal{E}}) \to j^m_* \Omega^\bullet_X(\widetilde{\mathcal{E}})$$

is a quasi-isomorphism.

Proof. Proposition 3.13 in [Del70]. □

7.3. **Regularity.**

7.3.1. *Regularity in dimension* 1. Let $\mathcal{O}$ be a discrete valuation ring of characteristic zero, with maximal ideal $\mathfrak{m}$ and field of fractions K. Suppose that $\mathcal{O}$ is endowed with a free rank-1 $\mathcal{O}$-module Ω, together with a derivation $\mathrm{d}\colon \mathcal{O} \to \Omega$, such that there exists a uniformizer t for which $\mathrm{d}t$ generates Ω.

7.3.2. *Regularity in dimension n.*

Theorem 7.3.1. Let X be a smooth complex algebraic variety. Then the functor $\mathcal{E} \mapsto \mathcal{E}^{an}$ gives an equivalence of categories between

(1) the category of algebraic locally free sheaf on X equipped with a regular integrable connection;
(2) the category of holomorphic locally free sheaf on X^{an} endowed with an integrable connection.

Proof. Theorem 5.9 in [Del70]. □

7.4. **Deligne's canonical extension.** Notations as above.

Definition 7.4.1. A local system $\mathcal{V}$ on X is said to be **unipotent along** D if the fundamental group $\pi_1(X)$ acts on this local system by unipotent transformations.

Definition 7.4.2. A flat bundle $(\mathcal{E},\nabla)$ on X is said to be **unipotent along** D if local system $\mathcal{E}^{\nabla=0}$ is unipotent along D.

Theorem 7.4.1 ([Del70, Proposition 5.2])**.** Let $(\mathcal{E},\nabla)$ be a flat bundle on X that is unipotent along D. Then there exists a unique extension $\widetilde{\mathcal{E}}$ of $\mathcal{E}$ on Y such that

(1) The matrix of the connection ∇, in an arbitrary local basis of $\widetilde{\mathcal{E}}$, has at worst a logarithmic pole along Y.
(2) The residue $\mathrm{Res}_i(\Gamma)$ of the connection along D_i is nilpotent.

Remark 7.4.1. In general, it's still possible to make an extension of $\mathcal{E}$. See Proposition 5.4 in [Del70].

7.5. **Proof of the theorem.**

Proposition 7.5.1 ([Del70, Lemma 6.9])**.** The map

$$\Omega_Y^\bullet(\log D)(\widetilde{\mathcal{E}}) \to j_*\Omega_X^\bullet(\mathcal{E})$$

is a quasi-isomorphism.

Part 3. **Appendix**

Appendix A. Resolutions of sheaves

A.1. **The acyclic sheaf.** In practice it may be difficult to choose an injective resolution, so we usually use other resolutions to compute sheaf cohomology.

Definition A.1.1. A sheaf $\mathcal{F}$ is **acyclic** if $H^i(X,\mathcal{F})=0$ for all $i>0$.

Example A.1.1. Every injective sheaf is acyclic.

Definition A.1.2. Let $\mathcal{F}$ be a sheaf. An **acyclic resolution** of $\mathcal{F}$ is an exact sequence

$$0\to\mathcal{F}\to\mathcal{A}^0\to\mathcal{A}^1\to\mathcal{A}^2\to\dots,$$

where $\mathcal{A}^i$ is acyclic for all i.

Proposition A.1.1. The cohomology of a sheaf can be computed using an acyclic resolution.

In fact, this is a standard homological technique called dimension shifting, so we state it in the language of homological algebra. Let us first see a simple version.

Example A.1.2. Let $\mathcal{F}$ be a left exact functor and let $0\to A\to M_1\to B\to 0$ be an exact sequence with M_1 $\mathcal{F}$-acyclic. Then $R^{i+1}\mathcal{F}(A)\cong R^i\mathcal{F}(B)$ for $i>0$, and $R^1\mathcal{F}(A)$ is the cokernel of $\mathcal{F}(M_1)\to\mathcal{F}(B)$.

Proof. By considering the long exact sequence induced by $0\to A\to M^1\to B\to 0$, one has

$$R^i\mathcal{F}(M^1)\to R^i\mathcal{F}(B)\to R^{i+1}\mathcal{F}(A)\to R^{i+1}\mathcal{F}(M^1).$$

(1) If $i>0$, then $R^i\mathcal{F}(M^1)=R^{i+1}\mathcal{F}(M^1)=0$ since M^1 is $\mathcal{F}$-acyclic, and thus $R^{i+1}\mathcal{F}(A)\cong R^i\mathcal{F}(B)$ for $i>0$.
(2) If $i=0$, then

$$0\to\mathcal{F}(M^1)\to\mathcal{F}(B)\to R^1\mathcal{F}(A)\to 0$$

implies $R^1\mathcal{F}(A)=\text{coker}\{\mathcal{F}(M^1)\to\mathcal{F}(B)\}$.

□

Now let's prove dimension shifting in a general setting.

Lemma A.1.1 (dimension shifting)**.** If

$$0\to A\to M^1\to M^2\to\cdots\to M^m\to B\to 0$$

is exact with each M^i $\mathcal{F}$-acyclic, then $R^{i+m}\mathcal{F}(A)\cong R^i\mathcal{F}(B)$ for $i>0$, and $R^m\mathcal{F}(A)$ is the cokernel of $\mathcal{F}(M^m)\to\mathcal{F}(B)$.

Proof. We prove it by induction on m. For $m=1$, we have already seen it in Example A.1.2. Assume it holds for $m<k$. For $m=k$, split $0\to A\to M^1\to M^2\to\cdots\to M^k\xrightarrow{\mathrm{d}_k} B\to 0$ into two exact sequences

$$0\to A\to M^1\to M^2\to\cdots\to M^{k-1}\to\ker \mathrm{d}_k\to 0$$

$$0\to \ker \mathrm{d}_k\to M^k\xrightarrow{\mathrm{d}_k} B\to 0.$$

Then by induction hypothesis, for $i>0$ we have

$$R^{i+k-1}\mathcal{F}(A)\cong R^i\mathcal{F}(\ker \mathrm{d}_k)$$
$$R^{i+1}\mathcal{F}(\ker \mathrm{d}_k)\cong R^i\mathcal{F}(B).$$

Combine these two isomorphisms together we obtain $R^{i+k}\mathcal{F}(A)\cong R^i\mathcal{F}(B)$ for $i>0$, as desired. For $i=0$, it suffices to let $i=1$ in $R^{i+k-1}\mathcal{F}(A)\cong R^i\mathcal{F}(\ker \mathrm{d}_k)$, then we obtain

$$R^k\mathcal{F}(A)=R^1\mathcal{F}(\ker \mathrm{d}_k)=\operatorname{coker}\{\mathcal{F}(M^k)\to\mathcal{F}(B)\}.$$

This completes the proof. □

Corollary A.1.1. If $0\to A\to M^\bullet$ is a $\mathcal{F}$-acyclic resolution, then $R^i\mathcal{F}(A)=H^i(\mathcal{F}(M^\bullet))$.

Proof. Truncate the resolution as

$$0\to A\to M^0\to M^1\to\ldots M^{i-1}\to B\to 0$$
$$0\to B\to M^i\to M^{i+1}\to\ldots$$

Since we already have $R^i\mathcal{F}(A)=\operatorname{coker}\{\mathcal{F}(M^{i-1})\to\mathcal{F}(B)\}$, and $\mathcal{F}$ is left exact, one has

$$\mathcal{F}(B)=\ker\{\mathcal{F}(M^i)\to\mathcal{F}(M^{i+1})\}.$$

Thus we obtain

$$R^i\mathcal{F}(A)=\operatorname{coker}\{\mathcal{F}(M^{i-1})\to\ker\{\mathcal{F}(M^i)\to\mathcal{F}(M^{i+1})\}\}=H^i(\mathcal{F}(M^\bullet)).$$

□

A.2. **The flabby sheaf.** The first kind of acyclic sheaf we introduce is the flabby sheaf[8].

Definition A.2.1. A sheaf $\mathcal{F}$ is **flabby** if for all open $U\subseteq V$, the restriction map $\mathcal{F}(V)\to\mathcal{F}(U)$ is surjective.

Example A.2.1. A constant sheaf on an irreducible topological space is flabby.

[8] Some authors also call this a flasque sheaf.

Proof. Note that the constant presheaf on an irreducible topological space is in fact a sheaf, and it is easy to see that the constant presheaf is flabby. □

In particular, we have

Example A.2.2. Let X be an algebraic variety. Then constant sheaf $\mathbb{Z}_X$ is flabby.

Example A.2.3. If $\mathcal{F}$ is a flabby sheaf on X, and $f: X \to Y$ is a continuous map, then $f_*\mathcal{F}$ is a flabby sheaf on Y.

Proof. For $V \subseteq W$ in Y, it suffices to show $f_*\mathcal{F}(W) \to f_*\mathcal{F}(V)$ is surjective, and that's exactly

$$\mathcal{F}(f^{-1}W) \to \mathcal{F}(f^{-1}V).$$

This is surjective since $\mathcal{F}$ is flabby. □

Example A.2.4. An injective sheaf is flabby.

Proof. Let $\mathcal{I}$ be an injective sheaf and $V \subseteq U$ be open subsets. Now we define sheaf $\underline{\mathbb{Z}}_U$ on X by

$$\underline{\mathbb{Z}}_U := \begin{cases} \underline{\mathbb{Z}}(W), & W \subseteq U \\ 0, & \text{otherwise} \end{cases}$$

where $\underline{\mathbb{Z}}$ is the constant sheaf valued in $\mathbb{Z}$, and similarly we define the sheaf $\underline{\mathbb{Z}}_V$. By construction, one has $\underline{\mathbb{Z}}_U(W) = \underline{\mathbb{Z}}_V(W)$ unless $W \subseteq U$ and $W \not\subseteq V$. Thus we obtain an exact sequence

$$0 \to \underline{\mathbb{Z}}_V \to \underline{\mathbb{Z}}_U.$$

Applying the functor $\mathrm{Hom}(-,\mathcal{I})$, which is exact, we obtain an exact sequence

$$\mathrm{Hom}(\underline{\mathbb{Z}}_U,\mathcal{I}) \to \mathrm{Hom}(\underline{\mathbb{Z}}_V,\mathcal{I}) \to 0.$$

Now let's explain why we need such a weird sheaf $\underline{\mathbb{Z}}_U$. In fact, we will prove $\mathrm{Hom}(\underline{\mathbb{Z}}_U,\mathcal{I}) = \mathcal{I}(U)$. Indeed since $\varphi\colon \underline{\mathbb{Z}}_U \to \mathcal{I}$ is a sheaf morphism. Then if $W \not\subseteq U$, then $\varphi(U)$ must be zero. If $W = U$, then the group of sections of $\underline{\mathbb{Z}}_U(U)$ over any connected component is simply $\mathbb{Z}$ and hence $\varphi(U)$ on this connected component is determined by the image of $1 \in \mathbb{Z}$. Thus $\varphi(U)$ can be thought of an element of $\mathcal{I}(U)$. Now on any proper open subset of U, φ is determined by restriction maps. Hence $\mathrm{Hom}(\underline{\mathbb{Z}}_U,\mathcal{I}) = \mathcal{I}(U)$, as desired. The same argument shows $\mathrm{Hom}(\underline{\mathbb{Z}}_U,\mathcal{I}) = \mathcal{I}(V)$, and thus we obtain an exact sequence

$$\mathcal{I}(U) \to \mathcal{I}(V) \to 0,$$

which shows $\mathcal{I}$ is flabby. □

Our goal is to prove a flabby sheaf is acyclic, but we still need some property of flabby sheaves.

Proposition A.2.1. If $0 \to \mathcal{F}' \xrightarrow{\phi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \to 0$ is an exact sequence of sheaves, and $\mathcal{F}'$ is flabby, then for any open subset U, the sequence

$$0 \to \mathcal{F}'(U) \xrightarrow{\phi(U)} \mathcal{F}(U) \xrightarrow{\psi(U)} \mathcal{F}''(U) \to 0$$

is exact.

Proof. It suffices to show $\mathcal{F}(U) \to \mathcal{F}''(U) \to 0$ is exact. Here we only gives a sketch of the proof. Since we have exact sequence on stalks for each $p \in U$ as follows

$$0 \to \mathcal{F}'_p \xrightarrow{\phi_p} \mathcal{F}_p \xrightarrow{\psi_p} \mathcal{F}''_p \to 0$$

Then for each $s \in \mathcal{F}''(U)$, there exists $t_p \in \mathcal{F}_p$ such that $\psi_p(t_p) = s|_p$, so there exists open subset $V_p \subseteq U$ containing p and $t \in \mathcal{F}(V_p)$ such that $\psi(t) = s|_{V_p}$. If we can glue these t together then we get a section in $\mathcal{F}(U)$ and is mapped to s, which completes the proof. However, they may not equal on the intersection. But things are not too bad, consider another point q and $t' \in \mathcal{F}(V_q)$ such that $\psi(t') = s|_{V_q}$, $(t'-t)|_{V_p \cap V_q} \in \ker\psi(V_p \cap V_q) = \operatorname{im}\phi(V_p \cap V_q)$. So there exists $t'' \in \mathcal{F}'(V_p \cap V_q)$ such that

$$\phi(t'') = (t'-t)|_{V_p \cap V_q}$$

Now since $\mathcal{F}'$ is flabby, then there exists $t''' \in \mathcal{F}(V_p)$ such that $t'''|_{V_p \cap V_q} = t''$. And consider $t + \phi(t''') \in \mathcal{F}(V_p)$, which will coincide with t' on $V_p \cap V_q$. After above corrections, we can glue t after correction together. $\square$

Proposition A.2.2. If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence of sheaves, and if $\mathcal{F}'$ and $\mathcal{F}$ are flabby, then $\mathcal{F}''$ is flabby.

Proof. Take $V \subseteq U$ and consider the following diagram

$$\begin{tikzcd}
0 \arrow[r] \arrow[d] & \mathcal{F}'(U) \arrow[r] \arrow[d] & \mathcal{F}(U) \arrow[r] \arrow[d] & \mathcal{F}''(U) \arrow[r] \arrow[d, dashed] & 0 \arrow[d] \\
0 \arrow[r] & \mathcal{F}'(V) \arrow[r] & \mathcal{F}(V) \arrow[r] & \mathcal{F}''(V) \arrow[r] & 0
\end{tikzcd}$$

Then the desired result follows from five lemma. $\square$

Proposition A.2.3. A flabby sheaf is acyclic.

Proof. Let $\mathcal{F}$ be a flabby sheaf. Since there are enough injective objects in the category of sheaf of abelian groups, there is an exact sequence

$$0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$$

with $\mathcal{I}$ injective. By Example A.2.4, $\mathcal{I}$ is flabby, and thus by Proposition A.2.2, $\mathcal{Q}$ is flabby. Consider the long exact sequence induced from the above short exact sequence:

$$\mathcal{F}(X) \to \mathcal{I}(X) \to \mathcal{Q}(X) \to H^1(X,\mathcal{F}) \to H^1(X,\mathcal{I}) \to \dots$$

Note that $H^1(X,\mathcal{I}) = 0$ since $\mathcal{I}$ is injective, and thus acyclic. Then $H^1(X,\mathcal{F}) = \operatorname{coker}\{\mathcal{I}(X) \to \mathcal{Q}(X)\}$. But Proposition A.2.1 shows that $\mathcal{I}(X) \to \mathcal{Q}(X)$ is surjective since $\mathcal{F}$ is flabby, so $H^1(X,\mathcal{F}) = 0$.

Now let us prove $H^k(X,\mathcal{F}) = 0$ for $k > 0$ by induction on k; the above argument shows that this is true for $k = 1$. Assume this holds for $k < n$, and consider

$$\dots \to H^{n-1}(X,\mathcal{Q}) \to H^n(X,\mathcal{F}) \to H^n(X,\mathcal{I}) \to H^n(X,\mathcal{Q}) \to \dots$$

By the induction hypothesis, we can reduce the above sequence to

$$\dots \to 0 \to H^n(X,\mathcal{F}) \to 0 \to H^n(X,\mathcal{Q}) \to \dots$$

which implies $H^n(X,\mathcal{F}) = 0$. This completes the proof. $\square$

A.3. **The soft sheaf.** The second kind of acyclic sheaf is the soft sheaf, which is quite similar to the flabby sheaf.

Definition A.3.1. A sheaf $\mathcal{F}$ over X is **soft** if for any closed subset $S \subseteq X$ the restriction map $\mathcal{F}(X) \to \mathcal{F}(S)$ is surjective.

Remark A.3.1. For closed subset S, the section over it is defined by

$$\mathcal{F}(S) := \varinjlim_{S \subseteq U} \mathcal{F}(U)$$

Parallel to Proposition A.2.1 and Proposition A.2.2, soft sheaves have the following properties:

Proposition A.3.1. If $0 \to \mathcal{F}' \xrightarrow{\phi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \to 0$ is an exact sequence of sheaves, and $\mathcal{F}'$ is soft, then the following sequence

$$0 \to \mathcal{F}'(X) \xrightarrow{\phi(X)} \mathcal{F}(X) \xrightarrow{\psi(X)} \mathcal{F}''(X) \to 0$$

is exact.

Proposition A.3.2. If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence of sheaves, and if $\mathcal{F}'$ and $\mathcal{F}$ are soft, then $\mathcal{F}''$ is soft.

Proposition A.3.3. A soft sheaf is acyclic.

One may wonder what the difference is between flabby and soft sheaves, since the definitions are quite similar and both kinds of sheaves are acyclic. By the definition of sections over a closed subset, every flabby sheaf is soft, but the converse fails.

Example A.3.1. The sheaf of smooth functions on a smooth manifold is soft but not flabby.

Lemma A.3.1. If $\mathcal{M}$ is a sheaf of modules over a soft sheaf of rings $\mathcal{R}$, then $\mathcal{M}$ is a soft sheaf.

Proof. Let $s \in \mathcal{M}(K)$ for some closed subset $K \subseteq X$. Then s extends to some open neighborhood U of K. Let $\rho \in \mathcal{R}(K \cup (X \setminus U))$ be defined by

$$\rho = \begin{cases} 1, & \text{on } K \\ 0, & \text{on } X \setminus U \end{cases}$$

Since $\mathcal{R}$ is soft, ρ extends to a section over X, and then $\rho \circ s$ is the desired extension of s. □

A.4. **The fine sheaf.** Another important kind of acyclic sheaf, which behaves like the sheaf of differential forms Ω^k_X, is called a fine sheaf. Recall the notion of a partition of unity: let $U = \{U_i\}_{i \in I}$ be a locally finite open cover of a topological space X. A partition of unity subordinate to U is a collection of continuous functions $f_i \colon U_i \to [0,1]$ for each $i \in I$ such that its support lies in U_i, and for any $x \in X$

$$\sum_{i \in I} f_i(x) = 1.$$

Definition A.4.1. A **fine sheaf** $\mathcal{F}$ on X is a sheaf of $\mathcal{A}$-modules, where $\mathcal{A}$ is a sheaf of rings such that for every locally finite open cover $\{U_i\}_{i \in I}$ of X, there is a partition of unity

$$\sum_{i \in I} \rho_i = 1$$

where $\rho_i \in \mathcal{A}(X)$ and $\operatorname{supp}(\rho_i) \subseteq U_i$.

Remark A.4.1. For a sheaf $\mathcal{F}$ on X and a section $s \in \mathcal{F}(X)$, its support is defined as

$$\operatorname{supp}(s) := \overline{\{x \in X : s|_x \neq 0\}}.$$

Proposition A.4.1. A fine sheaf is acyclic.

Proof. Let $\mathcal{F}$ be a sheaf of $\mathcal{A}$-modules and a fine sheaf. Choose an injective resolution

$$0 \to \mathcal{F} \xrightarrow{\mathrm{d}} \mathcal{I}^0 \xrightarrow{\mathrm{d}} \mathcal{I}^0 \xrightarrow{\mathrm{d}} \mathcal{I}^1 \xrightarrow{\mathrm{d}} \dots$$

such that $\mathcal{I}^i$ are injective sheaves of $\mathcal{A}$-modules. Let $s \in \mathcal{I}^p(X)$ such that $\mathrm{d}s = 0$. Then by exactness of injective resolution we have X is covered by open subsets U_i such that for each i there is an element $t_i \in \mathcal{I}^{p-1}(U_i)$ such that $\mathrm{d}t_i = s|_{U_i}$. By passing to a refinement we may assume that the cover $\{U_i\}$ is locally finite. Let $\{\rho_i\}$ be a partition of unity subordinate to $\{U_i\}$. Then we have $t = \sum \rho_i t_i \in \mathcal{I}^{p-1}(X)$ such that $\mathrm{d}t = s$. This completes the proof. $\square$

Example A.4.1. Let M be a smooth manifold and $\pi: E \to M$ be a vector bundle. Then the sheaf of smooth sections of E is a $C^\infty(M)$-module sheaf, which is a fine sheaf. In particular, the sheaf of tangent bundle, sheaf of differential forms Ω_M and k-forms Ω_M^k are fine sheaves.

Remark A.4.2. As a consequence, the cohomology of the sheaf of differential k-forms, or of any vector bundle over a smooth manifold, is not very interesting. In the complex setting, however, something interesting happens. Let $(X, \mathcal{O}_X)$ be a complex manifold and let $\pi: E \to X$ be a holomorphic vector bundle. Then the sheaf of holomorphic sections of E is not a fine sheaf, since partitions of unity need not be holomorphic. Thus the cohomology of a holomorphic vector bundle need not be trivial, and this is what Dolbeault cohomology computes.

Lemma A.4.1. Any fine sheaf is soft.

Proof. Let $\mathcal{F}$ be a fine sheaf, let $S \subseteq X$ be closed, and let $s \in \mathcal{F}(S)$. Let $\{U_i\}$ be an open cover of S and let $s_i \in \mathcal{F}(U_i)$ be such that

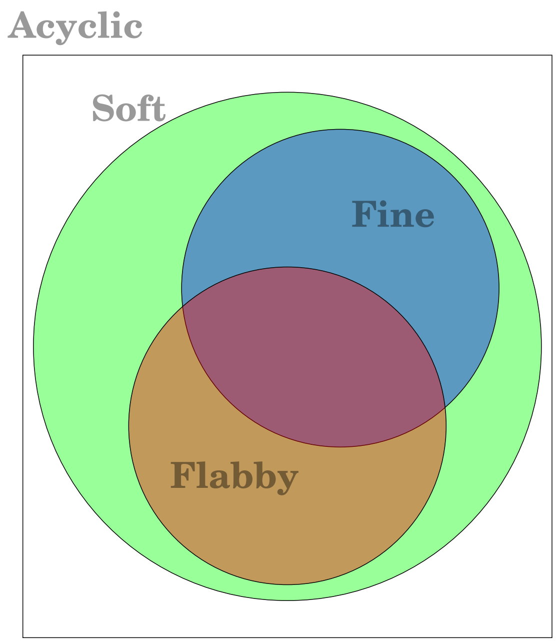
$$s_i|_{S\cap U_i} = s|_{S\cap U_i}.$$

Let $U_0 = X - S$, and let $s_0 = 0$. Then $\{U_i\} \coprod \{U_0\}$ is an open cover of X. Without loss of generality, we assume this open cover is locally finite and choose a partition of unity $\{\rho_i\}$ subordinate to it. Then

$$\overline{s} := \sum_i \rho_i(s_i)$$

is a section in $\mathcal{F}(X)$ which extends s. $\square$

Remark A.4.3. So far, we have shown that soft, fine, and flabby sheaves are acyclic. Lemma A.4.1 shows that a fine sheaf is soft, and by definition a flabby sheaf is soft. Example A.3.1 shows that a soft sheaf need not be flabby, and a constant sheaf on an irreducible space is flabby but not fine. In summary, we have the following relations:


Acyclic
Soft
Fine
Flabby

Appendix B. Local coordinates on algebraic varieties

B.1. **Smooth, unramified and étale morphism.**

B.1.1. *Smooth morphism.*

Definition B.1.1. Let $f: X \to Y$ be a morphism of schemes. It is said to be **locally of finite presentation** if for all $x \in X$, there exists an affine open neighborhood $x \in \operatorname{Spec} A \subseteq X$ and an affine open neighborhood $f(x) \in \operatorname{Spec} B \subseteq Y$ with the following property:

$$\begin{tikzcd} X \arrow[r, "f"] & Y \\ \operatorname{Spec} A \arrow[u, hook] \arrow[r] \arrow[d, hook] & \operatorname{Spec} B \arrow[u, hook] \\ \operatorname{Spec} B[t_1,\dots,t_n], \arrow[ur] & \end{tikzcd}$$

where $I = \ker(B[t_1,\dots,t_n] \to A)$ is finitely generated.

Definition B.1.2. A closed immersion $i: T \to \widetilde{T}$ between schemes is said to be **a first order thickening** if the ideal $\mathcal{I}$ of T in $\widetilde{T}$ satisfies $\mathcal{I}^2 = 0$.

Definition B.1.3. A first order thickening $T \hookrightarrow \widetilde{T}$ is said to be **affine first order thickening** if both T and $\widetilde{T}$ are affine.

Example B.1.1. Let R be a ring and $R[\epsilon] = R[x]/(x^2)$. Then $X_1 = \operatorname{Spec} R[\epsilon]$ is an affine first order thickening of $X = \operatorname{Spec} R$, where the closed immersion $X \hookrightarrow X_1$ is induced by the R-algebra morphism

$$\begin{aligned} R[\epsilon] &\to R \\ \epsilon &\mapsto 0. \end{aligned}$$

Definition B.1.4. Let $f: X \to S$ be a morphism of schemes. Then f is said to be **smooth** if

(1) f is locally of finite presentation;
(2) For every diagram

$$\begin{tikzcd} & & X \arrow[d, "f"] \\ T \arrow[urr, bend left] \arrow[r] & \widetilde{T} \arrow[r] & S, \end{tikzcd}$$

where $T \to \widetilde{T}$ is an affine first order thickening, there exists a morphism $\widetilde{T} \to X$ such that the following diagram commutes

$$\begin{tikzcd}
 & & X \arrow[d, "f"] \\
T \arrow[r] \arrow[urr, bend left] & \widetilde{T} \arrow[r] \arrow[ur, dashed] & S.
\end{tikzcd}$$

Remark B.1.1. A morphism is smooth if and only if $\mathrm{Hom}_S(\widetilde{T}, X) \to \mathrm{Hom}_S(T, X)$ is surjective for any affine thickening $T \hookrightarrow \widetilde{T}$.

Definition B.1.5. An S-scheme $X \to S$ is said to be **smooth** if the morphism $X \to S$ is smooth.

Proposition B.1.1.

(1) The smoothness is a local property.
(2) The smoothness is preserved by base change.

Example B.1.2. The projective space $\mathbb{P}^n_S$ is a smooth S-scheme.

Example B.1.3. Let X be an affine plane curve over a field k defined by $f \in k[x,y]$, that is, $X = \operatorname{Spec} k[x,y]/(f)$. Consider the following diagram

$$\begin{tikzcd}
 & & X \arrow[d] \\
\operatorname{Spec} k \arrow[r] \arrow[urr, bend left, "p"] & \operatorname{Spec} k[\epsilon] \arrow[r] \arrow[ur, dashed, "\theta"] & \operatorname{Spec} k.
\end{tikzcd}$$

Note that $p = (a,b)$ corresponding to a point in k^2 such that $f(a,b) = 0$. Then a lifting θ of p corresponds to a k-morphism

$$\theta^* : k[x,y]/(f) \to k[\epsilon]$$

such that $\theta^*(x) = a + u\epsilon, \theta^*(y) = b + v\epsilon$ with $u, v \in k^2$ satisfying

$$f(a + u\epsilon, b + v\epsilon) = 0.$$

Since $\epsilon^2 = 0$, we conclude from Taylor's formula that

$$f(a + u\epsilon, b + v\epsilon) = \left(\frac{\partial f}{\partial x}(a,b)u + \frac{\partial f}{\partial y}(a,b)v\right)\epsilon.$$

Thus, a lifting θ of $p = (a,b)$ corresponds to a pair $(u,v) \in k^2$ such that

$$\frac{\partial f}{\partial x}(a,b)u + \frac{\partial f}{\partial y}(a,b) = 0.$$

Intuitively, this is a tangent vector of X at p.

Remark B.1.2. Incidentally, if a morphism $\operatorname{Spec} k[\epsilon] \to S$ is to be seen as tangent vectors of S, then smoothness of a morphism $X \to S$ implies every tangent vector of S can be lifted to X, and that's how a "submersion" is defined in differential geometry.

Theorem B.1.1 ([Gro66b, 17.2.3])**.** If X is smooth S-scheme, then $\Omega^1_{X/S}$ is a locally free sheaf on X.

Theorem B.1.2 ([Gro66b, 17.2.3])**.** Let $\varphi\colon X \to Y$ is a S-morphism between S-schemes. If φ is smooth, then

$$0 \to \varphi^*\Omega^1_{Y/S} \to \Omega^1_{Y/S} \to \Omega^1_{X/Y} \to 0$$

is exact and locally split.

B.1.2. *Étale morphism.*

Definition B.1.6. Let $f: X \to S$ be a morphism of schemes. Then f is said to be **unramified** if

(1) f is locally of finite presentation;
(2) For every diagram

$$\begin{tikzcd} & & X \arrow[d, "f"] \\ T \arrow[r] \arrow[urr, bend left] & \widetilde{T} \arrow[r] & S, \end{tikzcd}$$

where $T \to \widetilde{T}$ is an affine first order thickening, there exists **at most one** morphism $\widetilde{T} \to X$ such that the following diagram commutes

$$\begin{tikzcd} & & X \arrow[d, "f"] \\ T \arrow[r] \arrow[urr, bend left] & \widetilde{T} \arrow[r] \arrow[ur, dashed] & S. \end{tikzcd}$$

Definition B.1.7. A morphism $f: X \to S$ is said to be **étale** if it is smooth and unramified.

Proposition B.1.2. Let $\varphi\colon X \to Y$ be a morphism of S-schemes locally of finite presentation and assume X is smooth over S. Then φ is étale if and only if $\varphi^*\Omega^1_{Y/S} \to \Omega^1_{X/S}$ is an isomorphism.

Proposition B.1.3. Let $\pi\colon X \to S$ be an étale morphism and $p \in X$ such that $\kappa_{\pi(p)} \cong \kappa_p$. Then the morphism of local rings $\mathcal{O}_{S,\pi(p)} \to \mathcal{O}_{X,p}$ induces an isomorphism

$$\widehat{\mathcal{O}}_{S,\pi(p)} \cong \widehat{\mathcal{O}}_{X,p}.$$

B.2. **Local coordinates on algebraic variety.**

Definition B.2.1. Let X be an S-scheme and $p \in X$. An **étale S-chart** of X at p is a family $(x_1,\dots,x_n)$ of sections of $\mathcal{O}_X$ in a neighborhood U of p such that the induced S-morphism

$$x = (x_1,\dots,x_n)\colon U \to \mathbb{A}^n_S$$

is étale.

Theorem B.2.1. Let X be a smooth S-scheme and $p \in X$. If $(x_1, \dots, x_n)$ is a family of sections of $\mathcal{O}_X$ in a neighborhood U of p, then

(1) $(x_1, \dots, x_n)$ defines an étale S-chart at p;
(2) $(\mathrm{d}x_1, \dots, \mathrm{d}x_n)$ trivializes $\Omega^1_{X/S}$ in a neighborhood p.

Corollary B.2.1. An S-scheme X is smooth if and only if every point of X admits an étale S-chart.

References

[Bry08] Jean-Luc Brylinski. *Loop spaces, characteristic classes and geometric quantization*. Modern Birkhäuser Classics. Birkhäuser Boston, Inc., Boston, MA, 2008. Reprint of the 1993 edition.

[Car53] Henri Cartan. Variétés analytiques complexes et cohomologie. In *Colloque sur les fonctions de plusieurs variables, tenu à Bruxelles, 1953*, pages 41–55. Georges Thone, Liège, 1953.

[Del70] Pierre Deligne. *Équations différentielles à points singuliers réguliers*, volume Vol. 163 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin-New York, 1970.

[DI87] Pierre Deligne and Luc Illusie. Relèvements modulo p^2 et décomposition du complexe de de Rham. *Invent. Math.*, 89(2):247–270, 1987.

[Gro66a] A. Grothendieck. On the de Rham cohomology of algebraic varieties. *Inst. Hautes Études Sci. Publ. Math.*, (29):95–103, 1966.

[Gro66b] Alexander Grothendieck. Éléments de géométrie algébrique: Iv. étude locale des schémas et des morphismes de schémas, troisième partie. *Publications Mathématiques de l'IHÉS*, 28:5–255, 1966.

[Har75] Robin Hartshorne. On the De Rham cohomology of algebraic varieties. *Inst. Hautes Études Sci. Publ. Math.*, (45):5–99, 1975.

[Har77] Robin Hartshorne. *Algebraic geometry*, volume No. 52 of *Graduate Texts in Mathematics*. Springer-Verlag, New York-Heidelberg, 1977.

[Hir64] Heisuke Hironaka. Resolution of singularities of an algebraic variety over a field of characteristic zero. I, II. *Ann. of Math. (2)*, 79:109–203; **79** (1964), 205–326, 1964.

[Mat86] Hideyuki Matsumura. *Commutative ring theory*, volume 8 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 1986. Translated from the Japanese by M. Reid.

[Ser55] Jean-Pierre Serre. Faisceaux algébriques cohérents. *Ann. of Math. (2)*, 61:197–278, 1955.

[Ser56] Jean-Pierre Serre. Géométrie algébrique et géométrie analytique. *Ann. Inst. Fourier (Grenoble)*, 6:1–42, 1955/56.

[Voi02] Claire Voisin. *Hodge theory and complex algebraic geometry. I*, volume 76 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2002. Translated from the French original by Leila Schneps.

[Wel80] R. O. Wells, Jr. *Differential analysis on complex manifolds*, volume 65 of *Graduate Texts in Mathematics*. Springer-Verlag, New York-Berlin, second edition, 1980.